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Exercise for osteoarthritis of the hip (Review)

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[Intervention Review]

Exercise for osteoarthritis of the hip

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ABSTRACT

Rationale

Hip osteoarthritis is a major public health issue. International treatment guidelines recommend exercise to manage symptoms. This is an update of a review first published in 2009 and last updated in 2014.

Objectives

To determine the effects of land-based exercise on pain, physical function, quality of life, participant-reported treatment success, study withdrawals, and adverse events in people with hip osteoarthritis.

Search methods

We searched CENTRAL, MEDLINE, Embase, and two trial registries from February 2013 (search date for last update) to 5 February 2025. We also checked reference lists for additional studies.

Eligibility criteria

We included randomised controlled trials (RCTs) of adults with hip osteoarthritis. Eligible comparisons were as follows.

- Exercise versus attention control/placebo.
- Exercise versus no treatment/usual care/limited education.
- Exercise plus a co-intervention (A) versus the co-intervention (A) alone.

We excluded perioperative exercise programmes; interventions involving vibration therapy, gait aids, or retraining; and studies that added unequal co-interventions in both groups or that compared only one form of exercise to another.

Outcomes

Our critical outcomes were pain, physical function, and quality of life. Our important outcomes were participant-reported treatment success, study withdrawals, and adverse events.

Exercise for osteoarthritis of the hip (Review)

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Risk of bias

As this was a review update, we used the original Cochrane risk of bias tool (RoB 1).

Synthesis methods

We used standard methodological procedures expected by Cochrane. We synthesised results for each outcome using random-effects meta-analysis where appropriate. We used the GRADE approach to assess certainty of evidence.

Included studies

Eighteen studies (1368 participants) met our inclusion criteria. Two studies (123 participants) evaluated exercise versus attention control or placebo, 10 studies (494 participants) evaluated exercise versus no treatment/usual care/limited education, and seven studies (751 participants) evaluated exercise plus a co-intervention (A) versus the co-intervention (A) alone. One trial (210 participants) included two comparator categories. Most studies were small and unblinded.

Synthesis of results

We converted all continuous effect estimates from standardised mean differences to mean differences (MDs) on a scale of 0 to 100. For pain and physical function, a negative MD indicates an improvement, while for quality of life, a positive MD indicates an improvement. Pain, physical function, quality of life, and participant-reported treatment success were measured immediately after treatment, while study withdrawals and adverse events were recorded at the end of follow-up.

Exercise versus attention control/placebo

Exercise, compared to attention control/placebo, may have little to no effect on pain (MD -6.31 points, 95% confidence interval (CI) -12.98 to 0.35; 2 studies, 123 participants; low certainty). Exercise may improve physical function slightly (MD -7.44 points, 95% CI -13.86 to -1.01; 2 studies, 123 participants; low certainty). No studies reported quality of life or participant-reported treatment success. Exercise may have little to no effect on study withdrawals (relative risk (RR) 0.83, 95% CI 0.23 to 3.03; 1 study, 106 participants; low certainty). Exercise may increase the risk of adverse events, but the evidence is very uncertain (RR 8.00, 95% CI 1.13 to 56.79; 1 study, 18 participants; very low certainty).

Exercise versus no treatment/usual care/limited education

Exercise, compared to no treatment/usual care/limited education, probably reduces pain slightly (MD -7.19 points, 95% CI -10.70 to -3.68; 9 studies, 449 participants; moderate certainty) and probably improves physical function slightly (MD -8.79 points, 95% CI -12.00 to -5.41; 9 studies, 447 participants; moderate certainty); however, these improvements are unlikely to be clinically meaningful. Exercise probably has little to no effect on quality of life (MD 2.31 points, 95% CI -1.15 to 5.91; 6 studies, 279 participants; moderate certainty). Exercise may have little to no effect on participant-reported treatment success (RR 1.57, 95% CI 0.62 to 3.99; 2 studies, 69 participants; low certainty) or study withdrawals (RR 1.51, 95% CI 0.80 to 2.86; 7 studies, 404 participants; low certainty). Exercise may have little to no effect on adverse events, but the evidence is very uncertain (RR 2.95, 95% CI 0.62 to 13.96; 6 studies, 257 participants; very low certainty).

Exercise plus a co-intervention (A) versus the co-intervention (A) alone

Exercise plus a co-intervention (A), compared to the co-intervention (A) alone, probably has little to no effect on pain (MD -3.86 points, 95% CI -8.07 to 0.35; 7 studies, 751 participants; moderate certainty), physical function (MD -2.37 points, 95% CI -6.59 to 1.86; 7 studies, 751 participants; moderate certainty), or quality of life (MD 3.60 points, 95% CI -1.30 to 8.36; 4 studies, 456 participants; moderate certainty). Exercise may have little to no effect on participant-reported treatment success (RR 1.25, 95% CI 0.62 to 2.54; 2 studies, 399 participants; low certainty), and probably has little to no effect on study withdrawals (RR 0.85, 95% CI 0.55 to 1.32; 6 studies, 683 participants; moderate certainty). Exercise probably reduces the risk of adverse events slightly (RR 0.75, 95% CI 0.58 to 0.97; 6 studies, 731 participants; moderate certainty).

Authors' conclusions

Compared with attention control or placebo, exercise may have little to no effect on pain and may improve physical function slightly. There was no evidence for quality of life or participant-reported treatment success. Exercise may have little to no effect on withdrawals. Evidence regarding adverse events is very uncertain.

Compared with no treatment, usual care or limited education, exercise probably improves pain and physical function slightly, although these effects are unlikely to be clinically meaningful. Exercise probably has little to no effect on quality of life and may have little to no effect on participant-reported treatment success or withdrawals. Evidence regarding adverse events is very uncertain.

Exercise plus a co-intervention (A), compared to the co-intervention (A) alone, probably has little to no effect on pain, physical function, quality of life, and study withdrawals, and may have little to no effect on participant-reported treatment success. Exercise probably reduces the risk of adverse events slightly.

Funding

No funding.

Registration

The original protocol was for a review on exercise for osteoarthritis of the hip or knee (<https://doi.org/10.1002/14651858.CD004376>). The review on hip osteoarthritis alone was first published in 2009 (<https://doi.org/10.1002/14651858.CD007912>), and last updated in 2014 (<https://doi.org/10.1002/14651858.CD007912.pub2>).

PLAIN LANGUAGE SUMMARY

Is exercise an effective therapy for hip osteoarthritis?

Key messages

- In people with hip osteoarthritis (a condition where the cartilage cushioning the hip joint wears down, causing pain and stiffness), exercise compared with dummy treatment may have little to no effect on pain, but may improve joint function slightly. Compared with no treatment, usual care, or limited education, exercise probably leads to small improvements in pain and joint function, although these may not be noticeable in daily life. Adding exercise to another treatment probably has little to no effect on pain and joint function.
- We are unsure if exercise leads to more unwanted events than dummy treatment or no treatment/usual care in people with hip osteoarthritis. When added to another treatment, exercise probably reduces the risk of unwanted events slightly.
- We need more and better-designed studies that focus specifically on people with hip osteoarthritis to draw more solid conclusions about the effects of exercise.

What is hip osteoarthritis?

Hip osteoarthritis occurs when the cartilage cushioning the hip joint wears down. It is a common, long-lasting condition. People with hip osteoarthritis often experience joint pain and stiffness. They may have difficulty performing everyday activities such as climbing stairs and getting up from a chair.

How might exercise help?

Exercise aims to improve muscle strength, joint movement, and self-belief in the capacity to perform certain activities. These effects should reduce pain and disability, helping people to better manage their symptoms. Exercise programmes vary considerably, with a wide range of different exercise types, intensities, and delivery methods.

What did we want to find out?

We wanted to know whether exercise reduces pain and improves joint function and well-being for people with hip osteoarthritis. We also wanted to know if people in the studies who exercised were more likely to report treatment success and less likely to drop out of the study or have unwanted events.

What did we do?

We searched for studies that assessed the effects of exercise in adults with hip osteoarthritis. Exercise programmes had to be land-based, planned, structured, and repetitive. We compared and summarised the results of the studies and rated our confidence in the evidence, based on factors such as study methods and how consistent the results were across the studies.

What did we find?

We found 18 studies with 1368 adults. Participants were mostly women (63%) and aged between 53 years and 74 years. Studies took place in Australia, Canada, Denmark, Finland, Germany, Ireland, Italy, the Netherlands, New Zealand, Norway, Sweden, and the USA. Exercise programmes lasted between two and 52 weeks. Most studies measured results immediately after the exercise programme; some studies also reported longer-term results.

Exercise compared with dummy treatment (2 studies, 123 adults)

- Exercise may have little to no effect on pain.
- Exercise may improve joint function slightly. On a scale of 0 to 100, where a 13-point difference is a meaningful change, people who exercised scored 7.4 points better than those on dummy treatment.
- No studies reported well-being or treatment success.
- People who exercise may be no more likely to stop treatment.
- We are unsure if there are more unwanted events with exercise.

Exercise compared with no treatment, usual care, or limited education (10 studies, 494 adults)

- Exercise probably improves pain and joint function, but only slightly. On a scale of 0 to 100, people who exercised scored 7.2 points better for pain (where a 12-point difference is a meaningful change) and 8.8 points better for joint function (where a 13-point difference is a meaningful change), compared with people receiving no treatment, usual care, or education.
- Exercise probably has little to no effect on well-being.
- People who exercise may be no more likely to report treatment success or to stop treatment.
- We are unsure if there are more unwanted events with exercise.

Exercise plus another treatment compared with the other treatment alone (7 studies, 751 adults)

- Exercise probably has little to no effect on pain, joint function, well-being, or whether people stop treatment.
- Exercise may have little to no effect on whether people report treatment success.
- People who exercise probably have slightly fewer unwanted events.

What are the limitations of the evidence?

Because the studies included people with different levels of hip pain and disability, the results are likely to apply to most adults with hip osteoarthritis. However, there were very few younger adults in the studies, so the findings may not apply as well to younger people.

In most studies, people knew whether they were receiving exercise, and pain and physical function were self-reported. This may make exercise appear more effective than it really is.

We are moderately confident in some of the results and less confident in others. This is because many studies were small, and the results varied across the studies. Further research may change the results.

How up to date is this evidence?

This review updates our previous review published in 2014. The evidence is up to date to February 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Exercise compared to attention control/placebo for hip osteoarthritis

Exercise compared to attention control/placebo for hip osteoarthritis

Patient or population: hip osteoarthritis

Setting: any non-perioperative setting in any country

Intervention: exercise

Comparison: attention control/placebo

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with attention control/placebo	Risk with exercise				
Pain score (immediately after treatment) follow-up: range 6 weeks to 3 months	The mean pain score was 25 points	mean 6.31 points fewer (12.98 fewer to 0.35 more)	-	123 (2 RCTs)	⊕⊕⊕⊕ Low ^{a,b,c,d}	Exercise may have little to no effect on pain compared with attention control/placebo. Based on a minimal clinically important difference of 12 points, the confidence interval of the mean difference included both a clinically important and clinically unimportant improvement. The standardised mean difference was -0.36 (95% CI -0.74 to 0.02).
Physical function score (immediately after treatment) follow-up: range 6 weeks to 3 months	The mean physical function score was 24.7 points	mean 7.44 points fewer (13.86 fewer to 1.01 fewer)	-	123 (2 RCTs)	⊕⊕⊕⊕ Low ^{a,b,d,e}	Exercise may improve physical function slightly compared with attention control/placebo. Based on a minimal clinically important difference of 13 points, the confidence interval of the mean difference included both a clinically important and clinically unimportant improvement. The standardised mean difference was -0.44 (95% CI -0.82 to -0.06).
Quality of life - not measured	-	-	-	-	-	No data identified for this outcome.
Participant-reported treatment success - not measured	-	-	-	-	-	No data identified for this outcome.

Study withdrawals (at final follow-up) follow-up: mean 3 months	100 per 1000	83 per 1000 (23 to 303)	RR 0.83 (0.23 to 3.03)	106 (1 RCT)	⊕⊕⊕⊕ Low ^{b,d,f}	Exercise may have little to no effect on study withdrawals compared with attention control/placebo.
Adverse events (at final follow-up) follow-up: mean 6 weeks	83 per 1000	667 per 1000 (94 to 1000)	RR 8.00 (1.13 to 56.79)	18 (1 RCT)	⊕⊕⊕⊕ Very low ^{b,d,g}	Exercise may increase adverse events compared with attention control/placebo, but the evidence is very uncertain.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; RR: risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_462268357814110886.

^a Standardised mean differences (SMDs) were back-translated to a 0–100 scale by multiplying a weighted average standard deviation (SD) calculated across all relevant study arms reporting outcomes on a 0–100 scale. The weighted SDs were 17.53 for pain and 16.90 for physical function. Predefined minimal clinically important differences were 12 points for pain (on a 0–100 scale where 0 is no pain), 13 points for physical function (on a 0–100 scale where 0 is no physical dysfunction), and 15 points for quality of life (on a 0–100 scale where 100 represents maximum quality of life). The risk with attention control/placebo was estimated using the mean from a representative study immediately after treatment: mean 25.0 (SD 16.2) for pain and mean 25.3 (SD 15.8) for physical function.

^b Downgraded one level for risk of bias as > 50% of trials were at high risk of performance and detection bias. While recognising that blinding of participants and personnel is difficult to achieve in these types of trials, failure to blind is problematic as trials with unblinded assessment of subjective outcomes are estimated to exaggerate the treatment benefit. Correcting for these biases would likely drive the improvements seen with exercise towards null.

^c Downgraded by one level for imprecision as total sample size was small and the confidence interval includes both no effect and a possible clinically important improvement (12 points).

^d Publication bias was not assessed due to limited studies (< 10).

^e Downgraded one level for imprecision, as total sample size was small, and the confidence interval includes a small effect and a possible clinically important improvement (13 points).

^f Downgraded one level for imprecision, as the low event number resulted in a wide confidence interval.

^g Downgraded by two levels for imprecision, as the low event number resulted in an extremely wide confidence interval.

Summary of findings 2. Summary of findings table - Exercise compared to no treatment/usual care/limited education for hip osteoarthritis

Exercise compared to no treatment/usual care/limited education for hip osteoarthritis

Patient or population: hip osteoarthritis

Setting: any non-perioperative setting in any country

Intervention: exercise

Comparison: no treatment/usual care/limited education

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with no treatment/usual care/limited education	Risk with exercise				
Pain score (immediately after treatment) follow-up: range 6 weeks to 3 months	The mean pain score was 27.0 points	mean 7.19 points lower (10.7 lower to 3.68 lower)	-	449 (9 RCTs)	⊕⊕⊕⊖ Moderate ^{a,b,c}	Exercise probably improves pain slightly compared with no treatment/usual care/limited education. However, the confidence interval of the mean did not meet the threshold for a clinically important improvement (12 points). The standardised mean difference was -0.41 (95% CI -0.61 to -0.21).
Physical function score (immediately after treatment) follow-up: range 6 weeks to 3 months	The mean physical function score was 25.3 points	mean 8.79 points lower (12 lower to 5.41 lower)	-	447 (9 RCTs)	⊕⊕⊕⊖ Moderate ^{a,b,c}	Exercise probably improves physical function slightly compared with no treatment/usual care/limited education. However, the confidence interval of the mean did not meet the threshold for a clinically important improvement (13 points). The standardised mean difference was -0.52 (95% CI -0.71 to -0.32).
Quality of life score (immediately after treatment) follow-up: range 8 weeks to 3 months	The mean quality of life score was 48.90 points	mean 2.31 points higher (1.15 lower to 5.91 higher)	-	279 (6 RCTs)	⊕⊕⊕⊖ Moderate ^{a,b,c}	Exercise probably has little to no effect on quality of life compared with no treatment/usual care/limited education. The confidence interval of the mean difference did not meet the threshold for a clinically important improvement (15 points). The standardised mean difference was 0.16 (95% CI -0.08 to 0.41).
Participant-reported treatment success (immediately after treatment)	222 per 1000	349 per 1000 (138 to 887)	RR 1.57 (0.62 to 3.99)	69 (2 RCTs)	⊕⊕⊖⊖ Low ^{b,c,d}	Exercise may have little to no effect on participant success compared with no treatment/usual care/limited education.

follow-up: range 8 weeks to 3 months						
Study withdrawals (at final follow-up) follow-up: range 8 weeks to 3 months	61 per 1000	92 per 1000 (49 to 174)	RR 1.51 (0.80 to 2.86)	404 (7 RCTs)	⊕⊕⊕⊕ Low ^{b,c,d}	Exercise may have little to no effect on study withdrawal compared with no treatment/usual care/limited education.
Adverse events (at final follow-up) follow-up: range 8 weeks to 12 months	19 per 1000	55 per 1000 (11 to 259)	RR 2.95 (0.62 to 13.96)	257 (6 RCTs)	⊕⊕⊕⊕ Very low ^{b,c,e}	Exercise may have little to no effect on adverse events compared with no treatment/usual care/limited education, but the evidence is very uncertain.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradeapro.org/presentations/#/isof/isof_question_revman_web_462268828527479697.

^a Standardised mean differences (SMDs) were back-translated to a 0–100 scale by multiplying a weighted average standard deviation (SD) calculated across all relevant study arms reporting outcomes on a 0–100 scale. The weighted SDs were 17.53 for pain, 16.90 for physical function, and 14.42 for quality of life. Predefined minimal clinically important differences were 12 points for pain (on a 0–100 scale where 0 is no pain), 13 points for physical function (on a 0–100 scale where 0 is no physical dysfunction), and 15 points for quality of life (on a 0–100 scale where 100 represents maximum quality of life). The risk with no treatment/usual care/limited education was estimated using the mean from a representative study immediately after treatment: mean 27.0 (SD 18.4) for pain, mean 24.7 (SD 15.8) for physical function, and mean 48.5 (SD 13.8) for quality of life.

^b Downgraded one level for risk of bias, as > 50% of trials were at high risk of performance and detection bias. While recognising that blinding of participants and personnel is difficult to achieve in these types of trials, failure to blind is problematic, as trials with unblinded assessment of subjective outcomes may exaggerate the treatment benefit. Correcting for these biases would likely drive the improvements seen with exercise towards null.

^c Publication bias was not assessed due to limited studies (< 10).

^d Downgraded one level for imprecision, as the low event number resulted in a wide confidence interval.

^e Downgraded two levels for imprecision, as the low event number resulted in an extremely wide confidence interval.

Summary of findings 3. Summary of findings table - exercise plus a co-intervention (A) vs. co-intervention (A) alone for hip osteoarthritis

Patient or population: hip osteoarthritis
Setting: any non-perioperative setting in any country
Intervention: exercise plus a co-intervention (A)
Comparison: co-intervention (A) alone

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with co-intervention (A) alone	Risk with exercise plus a co-intervention (A)				
Pain score (immediately after treatment) follow-up: range 2 weeks to 6 months	The mean pain score was 38 points	mean 3.86 points lower (8.07 lower to 0.35 higher)	-	751 (7 RCTs)	⊕⊕⊕⊖ Moderate ^{a,b,c}	Exercise plus a co-intervention (A) probably has little to no effect on pain compared with the co-intervention (A) alone. The confidence interval of the mean difference did not meet the threshold for a clinically important difference (12 points). The standardised mean difference was -0.22 (95% CI -0.46 to 0.02).
Physical function score (immediately after treatment) follow-up: range 2 weeks to 6 months	The mean physical function score was 35.7 points	mean 2.37 points lower (6.59 lower to 1.86 higher)	-	751 (7 RCTs)	⊕⊕⊕⊖ Moderate ^{a,b,c}	Exercise plus a co-intervention (A) probably has little to no effect on physical function compared with the co-intervention (A) alone. Based on a minimal clinically important difference of 13 points, the confidence interval of the mean difference included both a clinically important and clinically unimportant improvement. The standardised mean difference was -0.14 (95% CI -0.39 to 0.11).
Quality of life score (immediately after treatment) follow-up: range 2 weeks to 6 months	The mean quality of life score was 82.2 points	mean 3.6 points higher (1.3 lower to 8.36 higher)	-	456 (4 RCTs)	⊕⊕⊕⊖ Moderate ^{a,b,c}	Exercise plus a co-intervention (A) probably has little to no effect on quality of life compared with the co-intervention (A) alone immediately after treatment. The confidence interval of the mean difference did not meet the threshold for a clinically important difference (15 points). The SMD was 0.25 (95% CI -0.09 to 0.58).
Participant-reported treatment success (immediately after treatment) follow-up: mean 3 months	348 per 1000	435 per 1000 (216 to 885)	RR 1.25 (0.62 to 2.54)	399 (2 RCTs)	⊕⊕⊖⊖ Low ^{b,c,d}	Exercise plus a co-intervention (A) may have little to no effect on participant-reported success compared to the co-intervention (A) alone.

Study withdrawals (at final follow-up) follow-up: range 2 weeks to 6 months	106 per 1000	90 per 1000 (58 to 139)	RR 0.85 (0.55 to 1.32)	683 (6 RCTs)	⊕⊕⊕⊖ Moderate ^{b,c}	Exercise plus a co-intervention (A) probably has little to no effect on study withdrawals compared to the co-intervention (A) alone.
Adverse events (at final follow-up) follow-up: range 3 months to 6 months	242 per 1000	181 per 1000 (140 to 235)	RR 0.75 (0.58 to 0.97)	731 (6 RCTs)	⊕⊕⊕⊖ Moderate ^{b,c}	Exercise plus a co-intervention (A) probably reduces the risk of adverse events slightly compared to the co-intervention (A) alone.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

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^b Downgraded one level for risk of bias, as > 50% of trials were at high risk of performance and detection bias. While recognising that blinding of participants and personnel is difficult to achieve in these types of trials, failure to blind is problematic, as trials with unblinded assessment of subjective outcomes may exaggerate the treatment benefit. Correcting for these biases would likely drive the improvements seen with exercise towards null.

^c Publication bias was not assessed due to limited studies (< 10).

^d Downgraded one level for imprecision, as the low event number resulted in a wide confidence interval.

BACKGROUND

Description of the condition

Osteoarthritis is one of the most common forms of arthritis and is a leading contributor to global disability [1, 2]. The hip joint is among the most commonly affected joints, after the knee and hands [3]. In 2019, the global prevalence of hip osteoarthritis was approximately 5.5% [3], with a peak incidence in the age group of 60 to 64 years [4]. People with hip osteoarthritis typically experience pain, difficulty performing activities of daily living, and poor quality of life [2]. In addition, osteoarthritis is commonly associated with several comorbidities, such as metabolic syndrome and stroke [5]. Disease progression from onset of hip pain to end-stage disease is variable, and can lead to total hip arthroplasty [6]. For people with severe hip osteoarthritis who are candidates for surgery, total hip replacement provides significantly greater pain relief and functional improvement compared to exercise [7].

Diagnosis and classification of osteoarthritis may be based on radiographic findings, symptoms or clinical presentations, or self-report. However, there is considerable discordance between radiographic findings and symptoms in people with hip osteoarthritis [8, 9]. Clinical criteria include: age 45 years or over; joint pain during activity; and no morning joint stiffness, or morning joint stiffness lasting less than 30 minutes [10].

Description of the intervention and how it might work

There are several theoretical physical and psychosocial mechanisms by which exercise can improve pain and physical function in osteoarthritis [11]. Different types of exercise are thought to work through different mechanisms. For example, the beneficial effects of strengthening exercises may be due to improved lower limb muscle function, while the benefits of aerobic exercise may be due to improved cardiovascular fitness and increased proinflammatory cytokines [11]. There is evidence, albeit in knee osteoarthritis, that improvements in muscle strength, aerobic fitness, and inflammatory cytokines mediate symptom relief [12]. Irrespective of type, exercise may also improve psychosocial factors associated with osteoarthritis symptoms such as self-efficacy, fear of movement, and mental health [13, 14].

Why it is important to do this review

Exercise is a cornerstone treatment for hip osteoarthritis and is recommended as first-line management in current international guidelines [10, 15, 16]. This is an update of a Cochrane review first published in 2009 and last updated in 2014 (covering research published up to February 2013). The 2014 version included 10 trials and compared exercise with active comparators (defined as any non-exercise intervention) or placebo/no intervention/waiting list comparators [17]. With this update, we aimed to incorporate new randomised controlled trials (RCTs) investigating the effectiveness of exercise for hip osteoarthritis and to separate the analyses according to type of comparator (e.g. exercise versus attention control/placebo, exercise versus no treatment/usual care, and exercise plus a co-intervention versus the co-intervention alone). Comparators are important to consider given their impact on effect estimates [18]. We conducted this review update according to the Cochrane Musculoskeletal Editorial Board guidelines [19].

OBJECTIVES

To determine the effects of land-based exercise on pain, physical function, quality of life, participant-reported treatment success, study withdrawals, and adverse events in people with hip osteoarthritis.

METHODS

We followed the Methodological Expectations of Cochrane Intervention Reviews when conducting the review, and the Preferred Reporting Items for Systematic Reviews and Meta-Analysis for the reporting [20, 21].

Differences between protocol and review

- We did not perform a subgroup analysis according to delivery medium (i.e. in-person interventions with or without some remote component vs fully remote interventions), because few studies had fully remote interventions.
- We did not perform a subgroup analysis according to the number of exercise types prescribed (i.e. single type vs multiple types), because few studies had single-type exercise programmes.
- We did not perform meta-regression to assess the effect of the number of exercise sessions per week on outcomes, as fewer than 10 studies contributed to each comparator group [22].
- We did not assess publication bias, as fewer than 10 studies contributed to each comparator group [23].
- We performed a subgroup analysis by type of exercise (e.g. mind-body, mixed exercise types, resistance or neuromuscular, and aerobic), which was not planned in the protocol.

Differences between previous and current review update

- In this updated review, we returned to the comparator structure prespecified in the original protocol and classified comparators into three categories: exercise versus attention control/placebo, exercise versus no treatment/usual care/limited education, and exercise plus a co-intervention (A) versus the co-intervention (A) alone. Previous published versions combined comparators in the primary analyses. In this update, we conducted separate meta-analyses and prepared separate summary of findings tables for each comparator category to acknowledge that the effectiveness of exercise can depend on the comparator [18].
- We included additional outcomes (e.g. participant-reported treatment success and study withdrawals) to capture patient-important benefits and harms more comprehensively.
- We excluded quasi-randomised trials to improve methodological rigour and reduce the risk of bias.
- We added another time point of measurement (> 6 months) to evaluate the longer-term effects of exercise (beyond 3–6 months), as benefits from exercise for osteoarthritis may diminish over time, and shorter-term results do not fully capture long-term outcomes.
- We removed PEDro (Physiotherapy Evidence Database) and CINAHL (Cumulated Index in Nursing and Allied Health Literature) from the search strategy, because any relevant studies they include should already be captured in the other major databases that we searched (e.g. MEDLINE, Embase, the Cochrane Central Register of Controlled Trials (CENTRAL)).

- In this update, we explicitly stated that studies of people with osteoarthritis in other joints were eligible for inclusion if they presented the outcomes for people with hip osteoarthritis separately (or if we could obtain these data from the study authors), or if more than 80% of participants had hip osteoarthritis. The previous version of this review included studies with mixed populations (using data specific to hip osteoarthritis), but the methods section did not specify the inclusion criteria related to such studies.
- Because study participants could not be fully blinded to the treatment they received, we considered self-reported outcomes at high risk of detection bias, unless studies used limited disclosure procedures to minimise participants' awareness of the study aims and their group allocation, in which case we assigned a low-risk judgement. Limited disclosure procedures reduce expectancy effects and help ensure participants are unaware of hypothesised differences between groups, thereby limiting the potential for biased self-reporting.

Criteria for considering studies for this review

Types of studies

We included RCTs published in any language as full-text articles, protocols, secondary analyses, trial registry entries, and abstracts (e.g. conference abstracts), as well as unpublished RCTs.

Types of participants

We included adults (aged ≥ 18 years) with an established diagnosis of symptomatic hip osteoarthritis according to accepted clinical criteria [10, 24], or who self-reported hip osteoarthritis on the basis of chronic hip joint pain (with or without radiographic confirmation).

Trials of mixed populations (i.e. osteoarthritis in other joints as well as the hip) were eligible if they presented the outcomes for people with osteoarthritis of the hip separately (or if we could obtain separate data from the trial authors), or if more than 80% of participants had osteoarthritis of the hip.

Types of interventions

We included any land-based exercise regimens (defined as planned, structured, and repetitive activity). Exercise interventions eligible for inclusion could:

- incorporate education or behaviour change strategies (or both) designed to help participants adhere to the exercise programme (e.g. information about benefits of exercise, short message service (SMS) or mobile app to maximise adherence);
- be supervised or unsupervised; and
- include other non-surgical interventions alongside the exercise intervention, as long as they were permitted/provided similarly in any comparator groups.

Exercise interventions were ineligible if they:

- were perioperative (i.e. the study specifically recruited all participants from surgical waiting lists or evaluated outcomes after surgery);
- included whole-body vibration, gait aid prescription, or gait retraining as the intervention (although backward/retro walking

and sideways walking prescribed as a therapeutic exercise programme were eligible);

- included any other interventions (apart from education and behaviour change strategies) that were not equally included in any comparator groups (e.g. pain coping skills training); and
- were compared to another type of exercise.

Comparator (control) groups eligible for inclusion were:

- placebo, sham, or attention control (any intervention designed to control for contextual/placebo effects and described by the study authors as 'placebo' or 'sham', or designed to mimic the supportive elements of the primary intervention (care, attention, social interaction) and involving more than one bout of synchronous interaction with study personnel/care providers, other than contact to obtain outcome measures);
- no treatment (no defined intervention), usual care (not controlled by the trial), or minimal education/information (provided with a one-off information resource), and;
- any non-surgical intervention, including usual care, that was also offered/provided in the exercise group (e.g. exercise and diet vs diet only).

Outcome measures

Critical outcomes

In accordance with international consensus regarding the core set of outcome measures for phase III clinical trials in osteoarthritis [25], each trial had to assess at least one of the following.

- Hip pain severity.
- Physical function.
- Quality of life.

Studies were eligible if they met the design, population, and intervention criteria; however, we excluded trials that measured none of our critical outcomes, as they could not contribute data to this updated review. We were interested in the effect of exercise on our critical outcomes at three time points: immediately after treatment, three to six months after cessation of treatment, and more than six months after cessation of treatment. Where studies reported two or more eligible time points (in the 3–6-month and the > 6-month ranges), we used the latest time point. Each study was required to report measurement of outcomes for at least one of these time periods.

Important outcomes

- Participant-reported intervention success after treatment, as defined by the trialists
- Number of study dropouts/withdrawals at final follow-up
- Number and nature of adverse events (non-serious or serious in nature, deemed by the trial authors to be related to the treatment), as well as whether withdrawals from adverse events were reported, at final follow-up

Search methods for identification of studies

Electronic searches

We searched the reference list of the previous Cochrane review for relevant trials. We then conducted a new search from the end

search date of the previous review (February 2013) to 5 February 2025. We searched the following databases.

- MEDLINE OvidSP (February 2013 to 5 February 2025).
- Embase OvidSP (February 2013 to 5 February 2025).
- CENTRAL (2025, Issue 2) via The Cochrane Library (February 2013 to 5 February 2025).

We also searched ClinicalTrials.gov (<https://clinicaltrials.gov/>) for unpublished completed trials, and the World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP) search portal (<https://trialsearch.who.int/>) for ongoing trials, using the terms "hip osteoarthritis" and "exercise" (February 2013 to 5 February 2025).

We applied no language restrictions. [Supplementary material 1](#) shows the search strategy. Since the last version of this review, we updated the search terms for Embase based on the most recent Cochrane Highly Sensitive Search Strategy [26].

Searching other resources

We searched the reference lists of all included studies for any additional publications. In addition, we checked for postpublication amendments for all included and potentially eligible studies. We searched PubMed for retraction notices, errata, corrigenda, and expressions of concern to ensure that no included study had been retracted or updated in ways that would affect eligibility or interpretation.

Data collection and analysis

Selection of studies

Following deduplication and automatic exclusion of references in Covidence [27], four teams of two randomly paired review authors independently screened the titles and abstracts of all remaining references. All review authors participated in this process. We retrieved the full-text study reports/publications of any study that either of the two reviewers considered potentially eligible. Teams of two review authors independently screened the full-text reports to identify studies for inclusion, documenting reasons for exclusion of ineligible studies. Where any discrepancies arose, a third review author adjudicated. Review authors were not involved in screening trials that they had authored.

We collated multiple reports of the same study under a single reference ID so that each trial, rather than each report, was the unit of interest in the review. We recorded the selection process in sufficient detail to complete a PRISMA flow diagram [21]. Results of our search were managed in Covidence [27].

Data extraction and management

We extracted the following study characteristics and outcome data to a data collection form used in previous reviews [28].

- Methods: location, study setting, withdrawals, and date of study.
- Participants: number, mean age, sex, diagnostic criteria, important condition-specific baseline data (including body mass index (BMI), baseline pain, and physical function).
- Treatments: description of exercise treatment and comparator, including duration of treatment, setting, type/content of exercise, mode of treatment delivery (supervised or not,

delivered in-person or remotely, individual or group), number of contact occasions with healthcare provider, number of exercise sessions, number of exercises, frequency of exercise programme, individual session duration, and adherence. We collected characteristics of the exercise treatment based on the Consensus on Exercise Reporting Template (CERT) criteria [29].

- Outcomes: outcomes collected (as listed in [Outcome measures](#)), scales used, and measurement time points.
- Information on study design and methods for the risk of bias assessment (see [Risk of bias assessment in included studies](#)).
- Notes: trial funding and notable declarations of interest of trial authors.
- Information needed for the GRADE assessment (e.g. baseline risk in the control group for key outcomes).

Teams of two randomly paired review authors independently extracted outcome data from included trials. Review authors did not extract data from any trials that they had authored. We extracted the means, standard deviations (SDs), and number of participants per group for continuous outcomes. We noted in the 'Characteristics of included studies' table if outcome data were not reported in a usable way and when data were transformed ([Supplementary material 2](#)). We resolved disagreements by consensus or by involving a third review author. One review author (MH) transferred data into the RevMan file [30]. We double-checked that data were entered correctly by comparing the data presented in the systematic review with individual study reports, and another review author (BJL) double-checked that data from new studies were entered correctly. If the same scale was used over different time periods (e.g. pain over past week and over past two weeks on a visual analogue scale (VAS)), we extracted the outcome closest to the past week.

If a trial reported more than one pain outcome measure, we applied the following hierarchy (used in the previous review update [17]).

- Global pain (VAS or numeric rating scale (NRS)).
- Pain on walking (VAS or NRS).
- Western Ontario and McMaster Universities Arthritis Index (WOMAC) or Hip Disability and Osteoarthritis Outcome Score (HOOS) pain subscale.
- Other composite pain scales (e.g. Medical Outcomes Study 36-Item Short-Form Health Survey (SF-36) bodily pain subscale, Arthritis Impact Measurement Scale (AIMS) pain subscale).
- Pain on activities other than walking (VAS or NRS).
- Rest pain or pain during the night (VAS or NRS).

If a trial reported more than one physical function outcome measure, we applied the following hierarchy, established in consultation with the Cochrane Musculoskeletal Group.

- WOMAC physical function subscale or HOOS activities of daily living subscale.
- Composite disability scores other than WOMAC (e.g. Harris Hip Score).
- Global disability score (VAS or NRS).
- Walking disability.
- Disability other than walking.
- WOMAC total score.
- Lequesne Osteoarthritis Index global score.

If a trial reported more than one quality of life outcome measure, we applied the following hierarchy (used in the previous review update [17]).

- SF-36 Mental Component Summary (MCS).
- Medical Outcomes Study 12-Item Short-Form Health Survey (SF-12) MCS.
- EuroQol Five-Dimension Questionnaire (EQ-5D).
- Sickness Impact Profile.
- Nottingham Health Profile.
- Other quality of life scales.

We collected participant-reported treatment success data according to the following hierarchy.

- Dichotomous measure.
- Categorical global rating of change scale.
- Continuous measure.

To prevent selective inclusion of data based on the results, we used the following predefined decision rules to select data from trials.

- Where trials reported outcomes at multiple time points, we extracted data from the time point furthest from randomisation within the time points of interest.
- Where trialists reported both final values and change-from-baseline values for the same outcome, we extracted final values.
- Where trialists reported both unadjusted and adjusted values for the same outcome, we extracted adjusted values.
- We extracted intention-to-treat data for outcomes assessing benefits (pain, function, quality of life, and participant-reported treatment success) when reported, or extracted the number of participants analysed at that time point if data were not available for missing participants. For outcomes assessing adverse events, we extracted data for participants who received the treatment they were randomised to.

Risk of bias assessment in included studies

Two review authors (MH, BJJ) independently assessed the risk of bias for each new study using the Cochrane risk of bias tool RoB 1 [31]. For a study authored by MH, HF completed the risk of

bias assessment. Disagreements were resolved by discussion. We assessed risk of bias according to the following domains.

- Random sequence generation.
 - Criteria for low risk of bias judgement: the investigators used a random/unpredictable assignment sequence (e.g. rolling a dice, drawing of ballots with study group labels from a dark bag, computer-generated random sequence, pre-ordered sealed envelopes, and pre-ordered list of treatment assignments).
- Allocation concealment.
 - Criteria for low risk of bias judgement: the person responsible for randomising participants had no influence on the assignment sequence or on the decision about eligibility of the participant. The randomisation schedule was concealed via sealed opaque envelopes or was stored in a locked folder/server.
- Blinding of participants and personnel.
 - Criteria for low risk of bias judgement: exercise and control groups were indistinguishable for participants, or limited disclosure was used to blind participants to group components and study hypotheses.
- Blinding of outcome assessment.
 - Criteria for low risk of bias judgement: if participants were blinded, the blinding procedure was adequate for outcome assessment.
- Incomplete outcome data.
 - Criteria for low risk of bias risk judgement: the study report provided reasons for all withdrawals/dropouts or exclusion of participants from the analysis, and the percentage of withdrawals and dropouts did not exceed 20%. All randomised participants were reported/analysed in the group they were allocated to, irrespective of non-compliance.
- Selective outcome reporting.
 - Criteria for low risk of bias risk judgement: all the results from the prespecified outcomes were adequately reported in the published report of the trial. This was determined by comparing the protocol/trial registry and the report.

We graded each potential source of bias as high, low, or unclear risk, and provided a justification for our judgement in the risk of bias table (Supplementary material 2). The figure generated by RoB 1 provides a summary of the judgements (Figure 1).

Figure 1. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias): Self-reported outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
Abbott 2013	+	+	-	-	+	+	?
Bendrik 2021	?	+	-	-	+	+	+
Chopp-Hurley 2017	+	+	-	-	+	?	+
de Sire 2023	?	?	-	+	+	?	+
Fernandes 2010	+	+	-	-	+	+	+
Foley 2003	+	+	-	-	+	?	?
Fransen 2007	+	+	-	-	+	+	+
French 2013	+	+	-	-	+	+	+
Hall 2025	+	+	-	+	+	+	+
Hopman-Rock 2000	?	?	-	-	?	?	+
Juhakoski 2011	+	+	-	-	+	?	+
Krauß 2014	+	+	-	-	+	-	+
Sandal 2019	+	+	-	-	+	+	+
Tak 2005	+	?	-	-	-	?	+
Teirlinck 2016	+	+	-	-	+	+	+
Thompson 2020	+	?	-	-	+	+	+
van Baar 1998	+	+	-	-	+	?	+

Figure 1. (Continued)

van Baar 1998	+	+	+	+	+	+	+
Weber 2024	+	+	+	+	+	+	+

Measures of treatment effect

We used RevMan to analyse the data [30]. For continuous data (pain, physical function, and quality of life), we calculated the standardised mean difference (SMD) with 95% confidence interval (CI). To re-express SMDs as mean differences (MDs) on a 0-to-100 scale, we multiplied the pooled SMD by a weighted average SD calculated across all relevant study arms using that scale (i.e. $SD_{\text{weighted}} = \sum n_i SD_i / \sum n_i SD$) [32]. Our calculations are available in the open-access research repository Zenodo (<https://doi.org/10.5281/zenodo.18719377>). Dichotomous data were analysed as risk ratios (RRs) with 95% CIs.

To interpret the clinical importance of results, we compared the MDs to established minimal clinically important differences (MCIDs) on 0-to-100 scales for pain, physical function, and quality of life. These represent the magnitude of changes in symptoms that are thought to be meaningful for people with osteoarthritis and are frequently used to aid clinical interpretation of exercise effects in osteoarthritis exercise interventions. We used MCIDs from a systematic review that established the most credible differences for pain, physical function, and quality of life based on 13 studies, including nine studies on knee osteoarthritis and four on mixed samples of hip and knee osteoarthritis [33]. The MCIDs were 12 points for pain, 13 points for physical function, and 15 points for quality of life.

For dichotomous outcomes, we calculated the absolute percent change from difference in the risks between the exercise and control group using GRADEpro GDT software [34]. For continuous outcomes, we estimated the control group risk using the mean value from a single representative study (with the most representative study population and a large weighting in the overall result in RevMan).

Unit of analysis issues

The unit of analysis was the participant. Where multiple arms were reported in a single trial, we included only the relevant arms. If a meta-analysis included more than one intervention arm from the same trial (e.g. Exercise A vs Exercise B vs no intervention), we pooled the intervention arms. If a meta-analysis included more than one control group (e.g. attention control and usual care), we split the exercise group participants between comparisons for continuous outcomes, and we split the events between comparisons for dichotomous outcomes, in accordance with Chapter 23 of the *Cochrane Handbook for Systematic Reviews of Interventions* [35].

Dealing with missing data

We contacted investigators as needed to verify key study characteristics and obtain missing numerical outcome data (e.g. when data were not available for participants with hip osteoarthritis in studies with mixed samples). If we received no response after two attempts, we listed the trial as awaiting

classification. For continuous outcomes, we calculated SMDs based on the number of participants randomised at baseline. Where possible, we calculated missing SDs from other statistics, such as standard errors, CIs, or P values. Where continuous outcome data were reported as median and interquartile range, we converted these to mean and SD using established methods for meta-analysis of continuous outcomes [36].

Reporting bias assessment

Although originally planned, we did not assess publication bias with Egger's test and inspection of funnel plots [37], because all meta-analyses included fewer than 10 studies. To assess outcome reporting bias, we checked trial protocols against published reports or registries. For studies published after 1 July 2005, we searched the WHO ICTRP (<https://www.who.int/tools/clinical-trials-registry-platform>) for the a-priori trial protocol.

Synthesis methods

We conducted meta-analyses using a random-effects model to account for expected clinical and methodological variability across studies.

Meta-analyses were conducted separately according to the following prespecified comparator categories.

- Exercise versus attention control or placebo.
- Exercise versus no treatment, usual care, or limited education.
- Exercise plus a co-intervention (A) versus the co-intervention (A) alone.

We did not pool studies across comparator categories, as effectiveness is known to differ according to comparator [18], and these comparisons address conceptually distinct clinical questions. Comparisons against no treatment/usual care/limited education estimate the effect of exercise instead of no treatment/usual care/limited education (substitution effect), whereas comparisons in which exercise is added to usual care or another active intervention estimate the incremental benefit of exercise beyond that intervention (additive effect).

Within each comparator category, outcomes were analysed separately according to follow-up time point. Pooled effect estimates were calculated within each comparator and follow-up category where sufficient data were available.

We prepared separate summary of findings tables for each comparator category, to ensure alignment between the analytic structure and presentation of results.

Investigation of heterogeneity and subgroup analysis

We evaluated clinical and methodological diversity in terms of participants, interventions, outcome measurement tools, and study characteristics, to confirm whether a meta-analysis was appropriate. To assess statistical heterogeneity, we visually

inspected the forest plots for obvious differences in results between studies, and we used the I^2 statistic and χ^2 test to assist interpretation.

We interpreted the I^2 values using the following rough thresholds [37].

- 0% to 40%: might not be important.
- 30% to 60%: may represent moderate heterogeneity.
- 50% to 90%: may represent substantial heterogeneity.
- 75% to 100%: may represent considerable heterogeneity.

We kept in mind that the observed value of I^2 depends on the magnitude and direction of effects and the strength of evidence for heterogeneity (e.g. P value from the χ^2 test, or a CI for the I^2 statistic: uncertainty in the value of the I^2 statistic is substantial when the number of studies is small).

We interpreted the χ^2 test results such that a P value of 0.10 or less indicated evidence of statistical heterogeneity. If we identified statistical heterogeneity, we reported this and investigated possible causes [37]. This included removing potential outliers with extreme results.

We performed analyses to assess if there were differences in pain, physical function, and quality of life according to type of exercise: aerobic; balance/proprioceptive; mind-body (e.g. yoga, tai chi); stretching; mixed types of exercises (e.g. programmes that include more than one type of exercise); and resistance, neuromuscular, or both.

To compare the effects of the different types of exercise programme, we used the formal test for subgroup interactions in RevMan and examined the magnitude of the effects [30].

Equity-related assessment

We did not investigate equity-related characteristics in this review.

Sensitivity analysis

We performed sensitivity analyses to interrogate the robustness of the effect of exercise on pain, physical function, and quality of life

immediately after treatment. We planned to assess the effect of excluding studies at high or unclear risk of selection bias, detection bias, and attrition bias. We also decided post hoc to assess the effect of combining change scores with final scores when using SMDs; this is usually not advised because the SDs used in the standardisation reflect different sources of variability [22].

Certainty of the evidence assessment

We created separate summary of findings tables for each comparison category using GRADEpro GDT software [34]. Our summary of findings tables included all our critical and important outcomes.

An independent assessor without any involvement in the studies (Dr Renea Johnston from the Cochrane Musculoskeletal Group) assessed the certainty of the evidence across all studies that contributed to the meta-analysis for each outcome, using the five GRADE considerations (risk of bias, consistency of effect, imprecision, indirectness, and publication bias), as outlined in the *Cochrane Handbook for Systematic Reviews of Interventions* [37]. The GRADE assessment was based on criteria used previously [38]. The certainty of evidence was reported as high, moderate, low, or very low. We justified, documented, and incorporated judgements into the reporting of results for each outcome.

Consumer involvement

Consumers were not involved in the update of this review.

RESULTS

Description of studies

Results of the search

The updated search identified 3892 records (3891 from electronic database searching and 1 from other sources; [Figure 2](#)). After automatic removal of 460 duplicates and 905 non-RCTs, we screened the titles and abstracts of the 2527 remaining records, excluding 2420 as clearly irrelevant. Of the 107 remaining records, we excluded 54 (52 studies) for reasons detailed in [Supplementary material 3](#).

Figure 2. Study flow diagram.

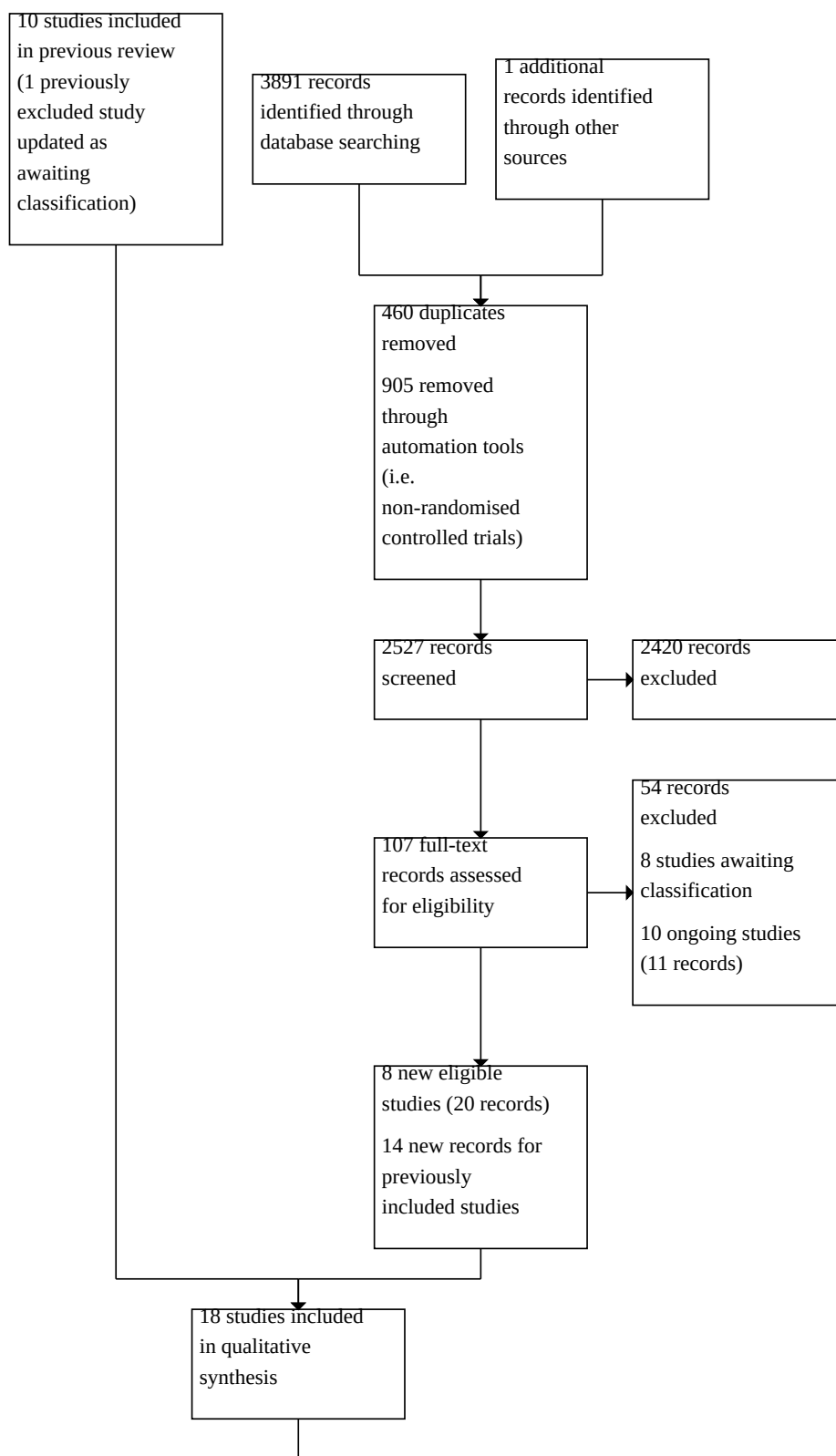
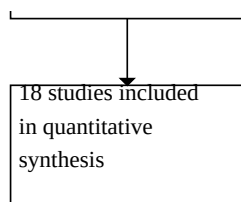


Figure 2. (Continued)



We included eight new trials (Bendrik 2021 [39, 40, 41]; Chopp-Hurley 2017 [42]; de Sire 2023 [43]; Hall 2025 [44, 45, 46]; Krauß 2014 [47, 48, 49, 50]; Sandal 2019 [51]; Teirlinck 2016 [52, 53, 54, 55, 56]; Weber 2024 [57, 58]). The previous version of this review included 10 trials (Abbott 2013 [59, 60, 61, 62, 63, 64, 65, 66]; Fernandes 2010 [67, 68, 69, 70, 71, 72, 73]; Foley 2003 [74]; Fransen 2007 [75]; French 2013 [76, 77, 78, 79, 80]; Hopman-Rock 2000 [81]; Juhakoski 2011 [82]; Tak 2005 [83]; Thompson 2020 [84, 85]; van Baar 1998 [86, 87]). One trial included as an abstract in the previous review was included in this update as a full-text article (Thompson 2020). One trial reported pain and function in two separate publications (Fernandes 2010). In total, this review update included 18 trials with 1368 participants. [Supplementary material 2](#) presents the characteristics of the included studies.

We listed nine studies as awaiting classification for different reasons: four studies were published as abstract only, and we were unable to confirm eligibility (Fugaru 2022 [88]; Jigami 2013 [89]; Matei 2014 [90]; Marcu 2013 [91]); one trial was marked as complete in the registry, but we were unable to find any published data (NCT03872869 [92]); and four trials reported no relevant data, and study authors did not respond to our requests for more information (Bossen 2013 [93]; Hinrichs 2016 [94]; Park 2017 [95]; Ravaud 2004 [96]). One study in the latter group had been excluded from the previous version of this review, but we now consider it potentially relevant (Ravaud 2004). [Supplementary material 4](#) presents the characteristics of studies awaiting classification. We identified records for 10 ongoing studies: five were protocols (Bennell 2023 [97]; Hastings 2024 [98]; Nicolson 2023 [99, 100]; Semciw 2018 [101]; Weber 2023 [102]), and five were registry records not marked as complete or recently marked as completed (DRKS00018847 [103]; NCT05272748 [104]; NCT06391554 [105]; UMIN ID UMIN00039723 [106]; U1111-1253-7874 [107]). [Supplementary material 5](#) presents the characteristics of ongoing studies.

We contacted 12 study authors to request additional data. Five authors provided the requested data (Bendrik 2021; Chopp-Hurley 2017; Fernandes 2010; Sandal 2019; Weber 2024), one author confirmed that the relevant data were already included in the published report (Krauß 2014), and one author confirmed that no separate data were available for people with hip osteoarthritis (Walsh 2020 [108, 109]). Four authors of studies awaiting classification did not respond to our requests for more information (Bossen 2013; Hinrichs 2016; Park 2017; Ravaud 2004), and one author of an included study did not respond to our request for longer-term data (Abbott 2013).

Included studies

[Table 1](#) presents a summary of the included trials.

Study design and setting

Trials were conducted in Australia (Foley 2003; Fransen 2007; Hall 2025), Canada (Chopp-Hurley 2017), Denmark (Sandal 2019), Finland (Juhakoski 2011), Germany (Krauß 2014; Weber 2024), Ireland (French 2013), Italy (de Sire 2023), the Netherlands (Hopman-Rock 2000; Tak 2005; Teirlinck 2016; van Baar 1998), New Zealand (Abbott 2013), Norway (Fernandes 2010), Sweden (Bendrik 2021), and the USA (Thompson 2020).

Participants

[Supplementary material 8](#) provides a summary of participant characteristics. The mean participant age ranged from 53 years to 73 years, and mean BMI ranged from 24.7 kg/m² to 30.5 kg/m². Diagnosis of hip osteoarthritis was based on the American College of Rheumatology criteria (Abbott 2013; Bendrik 2021; Chopp-Hurley 2017; de Sire 2023; Fransen 2007; French 2013; Hopman-Rock 2000; Juhakoski 2011; Krauß 2014; Tak 2005; Teirlinck 2016; van Baar 1998; Weber 2024), radiographic classification (Foley 2003), radiographic classification and symptoms (Fernandes 2010; Thompson 2020), and symptoms only (Hall 2025; Sandal 2019).

Interventions

[Table 2](#) summarises the characteristics of exercise interventions and comparators. A CERT checklist summary for each study is available in the research repository Zenodo (<https://doi.org/10.5281/zenodo.15253832>). The exercise interventions varied considerably in terms of type of exercise(s), training variables (e.g. frequency, intensity, and duration of sessions), and duration of the intervention ([Table 2](#)). All interventions were delivered in person.

The comparators were attention control or placebo in two studies (Foley 2003; Krauß 2014), no treatment/usual care/limited education in 10 studies (Abbott 2013; Chopp-Hurley 2017; Fransen 2007; French 2013; Hopman-Rock 2000; Krauß 2014; Sandal 2019; Tak 2005; Thompson 2020; Weber 2024), and a co-intervention (also used in the exercise group) in seven studies (Bendrik 2021; de Sire 2023; Fernandes 2010; Hall 2025; Juhakoski 2011; Teirlinck 2016; van Baar 1998). One study included comparators from two categories: no treatment/usual care/limited education and attention control/placebo (Krauß 2014).

Outcomes

[Table 3](#) summarises the outcome measures reported in each study. Follow-up ranged from two weeks to 24 months after randomisation.

All studies measured pain ([Table 3](#)). Seven studies measured global/overall pain on an NRS/VAS (de Sire 2023; Hall 2025; Hopman-

Rock 2000; Tak 2005; Teirlinck 2016; Thompson 2020; van Baar 1998), nine studies measured pain using the WOMAC/HOOS pain subscale (Abbott 2013; Bendrik 2021; Chopp-Hurley 2017; Foley 2003; Fransen 2007; Juhakoski 2011; Krauß 2014; Sandal 2019; Weber 2024), and two studies measured pain on activity using an NRS/VAS (Fernandes 2010; French 2013).

All studies measured physical function (Table 3). Fourteen studies used the WOMAC or HOOS activities of daily living subscale (Abbott 2013; Bendrik 2021; Chopp-Hurley 2017; Fernandes 2010; Foley 2003; Fransen 2007; French 2013; Hall 2025; Juhakoski 2011; Krauß 2014; Sandal 2019; Teirlinck 2016; Thompson 2020; Weber 2024), two used the Impact of Rheumatic Disease on General Health and Lifestyle scale (Hopman-Rock 2000; van Baar 1998), one used the Harris Hip Score (de Sire 2023), and one used the Groningen Activity Restriction Scale (Tak 2005).

Ten studies measured quality of life (Table 3). Two studies used the MCS score of the SF-12/36 (Fransen 2007; French 2013), three used the HOOS quality of life subscale (Chopp-Hurley 2017; Sandal 2019; Weber 2024), one used the Six-dimension Assessment of Quality of Life (AQoL-6D; Hall 2025), one used a generic scale (NRS or VAS; Tak 2005), and three used the EQ-5D (Bendrik 2021; de Sire 2023; Teirlinck 2016).

Four studies measured participant-reported treatment success (Hall 2025; Sandal 2019; Teirlinck 2016; Thompson 2020). One study defined response as more than a 13-mm improvement in pain on a 100-mm scale (Thompson 2020), two used global rating of change (Hall 2025; Sandal 2019), and one used global rating of recovery (Teirlinck 2016).

Thirteen studies measured adverse events (Table 3). Most adverse events were mild and musculoskeletal in nature (Table 4).

Excluded studies

We excluded 54 records after full-text assessment (Figure 2). Supplementary material 3 presents the characteristics of excluded studies and reasons for exclusion.

Risk of bias in included studies

Figure 1 summarises the risk of bias judgements for each study.

Allocation

We judged 12 studies at low risk of selection bias overall, because they adequately generated a random sequence and concealed it until allocation (Abbott 2013; Chopp-Hurley 2017; Fernandes 2010; Foley 2003; Fransen 2007; French 2013; Hall 2025; Juhakoski 2011; Krauß 2014; Sandal 2019; Teirlinck 2016; van Baar 1998). Three studies did not report their methods of random sequence generation, so were at unclear risk for this aspect of selection bias (Bendrik 2021; de Sire 2023; Hopman-Rock 2000). The allocation concealment method was unclear in five studies (de Sire 2023; Hopman-Rock 2000; Tak 2005; Thompson 2020; Weber 2024).

Blinding

We considered all studies at high risk of performance bias. Outcomes were self-reported, and we rated two studies at low risk of detection bias because they adequately blinded participants (de

Sire 2023; Hall 2025). The remaining trials were deemed at high risk of detection bias.

Incomplete outcome data

The risk of attrition bias was considered low in 15 studies, because dropouts were low (< 20%) and consistent across interventions, or because the study used intention-to-treat analyses (Abbott 2013; Bendrik 2021; Chopp-Hurley 2017; de Sire 2023; Fernandes 2010; Foley 2003; Fransen 2007; French 2013; Hall 2025; Juhakoski 2011; Krauß 2014; Sandal 2019; Teirlinck 2016; Thompson 2020; van Baar 1998). We considered Hopman-Rock 2000 at unclear risk and Tak 2005 and Weber 2024 at high risk of attrition bias.

Selective reporting

We considered 10 studies at low risk of reporting bias because they were prospectively registered, and they reported results for all outcomes in the trial registry (Abbott 2013; Bendrik 2021; Fernandes 2010; Fransen 2007; French 2013; Hall 2025; Sandal 2019; Teirlinck 2016; Thompson 2020; Weber 2024). Reporting bias was unclear in seven studies, as they were not registered, or we were unable to access the trial registry (Chopp-Hurley 2017; de Sire 2023; Foley 2003; Hopman-Rock 2000; Juhakoski 2011; Tak 2005; van Baar 1998). Krauß 2014 was registered but did not report all outcomes described in the trial registry (high risk of bias).

Other potential sources of bias

Abbott 2013 had an eight-month interval between the end of monitored treatment and outcome assessment, and some participants in Foley 2003 were on an orthopaedic waiting list. We rated these two studies at unclear risk of other bias, and the remaining studies at low risk.

Synthesis of results

Supplementary material 6 presents all analyses.

We back-translated all SMDs to MDs on a scale of 0 to 100 points. For pain and physical function, 0 represents no pain and no physical dysfunction (i.e. a negative MD indicates an improvement); whereas for quality of life, 100 represents the best possible quality of life (i.e. a positive MD indicates an improvement).

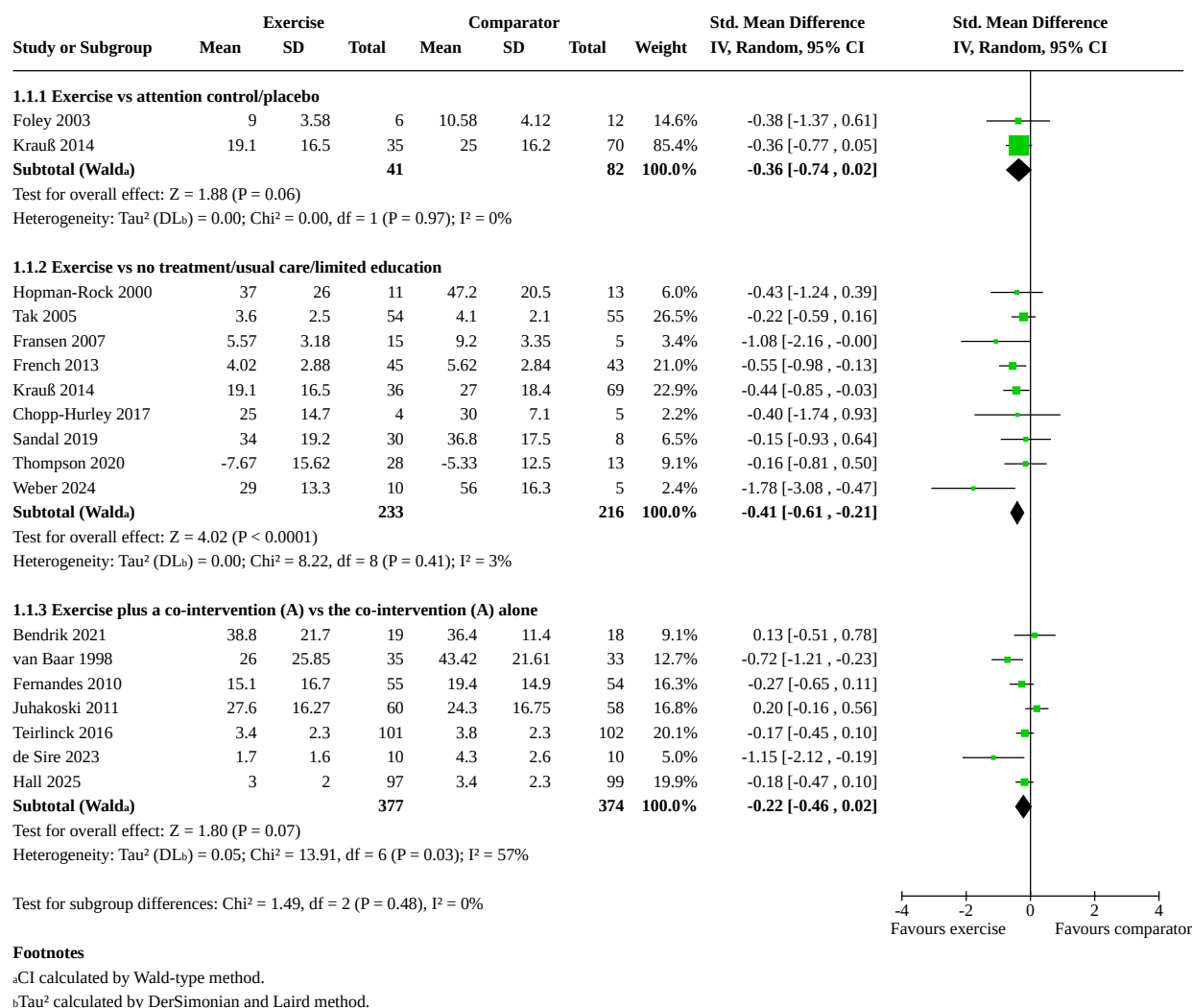
Exercise versus placebo/attention control

Two studies (123 participants) assessed exercise compared with placebo or attention control (Foley 2003; Krauß 2014). Summary of findings 1 presents the primary analysis results.

Pain

Exercise may have little to no effect on hip pain immediately after treatment compared to placebo/attention control (SMD -0.36, 95% CI -0.74 to 0.02; 2 studies, 123 participants; Figure 3; Analysis 1.1; low-certainty evidence). These data represent a mean improvement of 6.31 points with exercise (MD -6.31 points, 95% CI -12.98 to 0.35). The CI of the MD included both no effect and a possible clinically important improvement (12 points). We downgraded the evidence certainty one level for high risk of bias (both studies were at high risk of performance and detection bias) and one level for imprecision (few participants and a CI including both no effect and a possible clinically important improvement).

Figure 3. Effect on pain immediately after treatment.



Sensitivity analyses

Sensitivity analyses were not feasible. Both studies in the primary analysis were at low risk of selection and attrition bias, and neither study was at low risk of detection bias (Foley 2003; Krauβ 2014).

Sustainability of effects

No studies reported pain at three to six months after treatment or more than six months after treatment.

Subgroup analysis: type of exercise

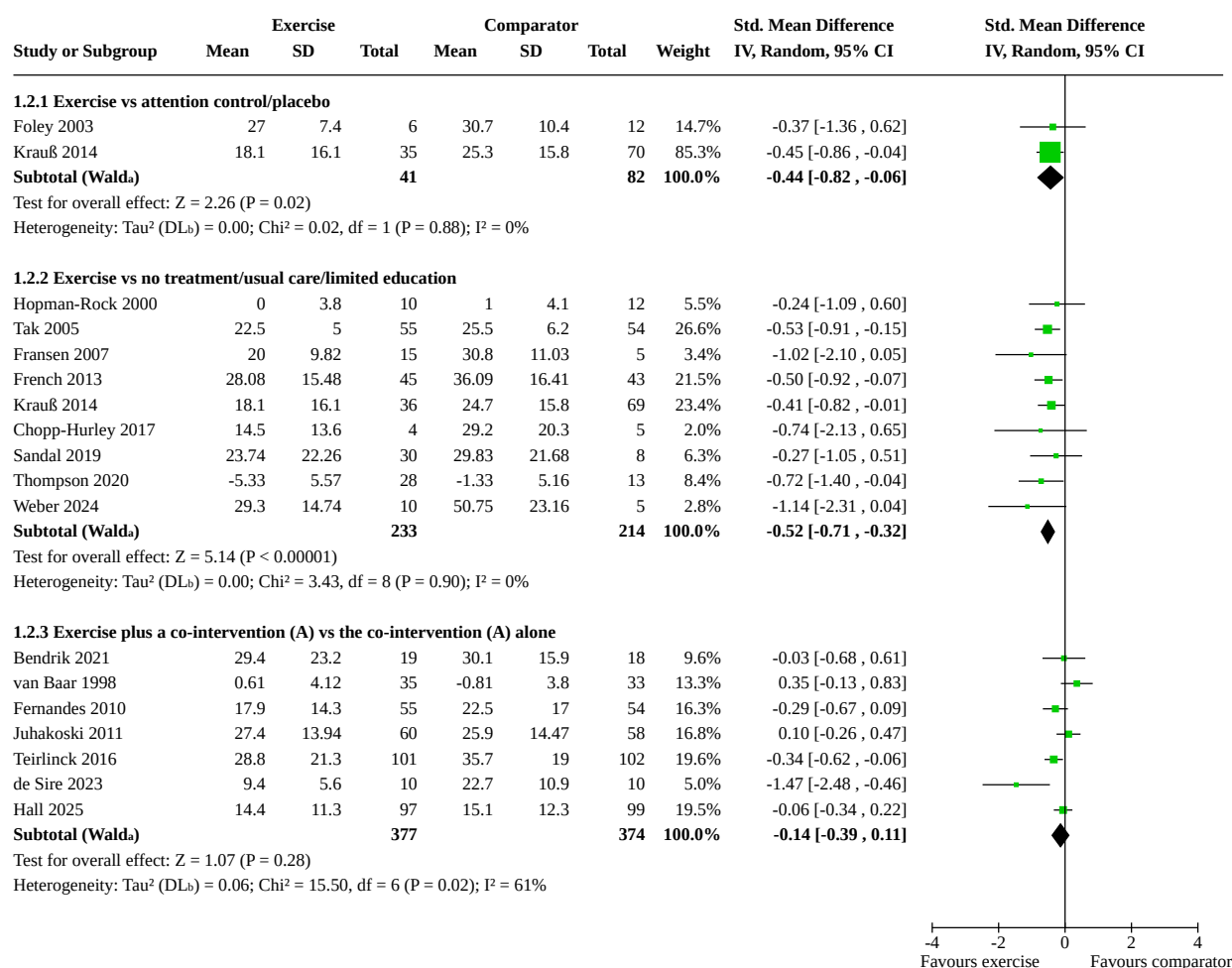
One study (105 participants) evaluated mixed exercise (Krauβ 2014); and one study (18 participants) evaluated resistance training, neuromuscular exercise, or both (Foley 2003). There was no

evidence of a difference in effect according to type of exercise (test for subgroup differences: $P = 0.97$, $I^2 = 0\%$; Analysis 5.1).

Physical function

Exercise may improve physical function immediately after treatment compared to placebo/attention control (SMD -0.44, 95% CI -0.82 to -0.06; $I^2 = 0\%$; 2 studies, 123 participants; Figure 4; Analysis 1.2; low-certainty evidence). These data represent a mean improvement of 7.44 points with exercise (MD -7.44 points, 95% CI -13.86 to -1.01). The CI of the MD crossed the threshold for a clinically important improvement (13 points). We downgraded the evidence certainty one level for high risk of bias (> 50% of included studies at high risk of performance and detection bias), and one level for imprecision (few participants and a CI including both no effect and a possible clinically important improvement).

Figure 4. Effect on physical function immediately after treatment.



Footnotes

^aCI calculated by Wald-type method.^b Tau^2 calculated by DerSimonian and Laird method.

Sensitivity analyses

Sensitivity analyses were not feasible. Both studies in the primary analysis were at low risk of selection and attrition bias, and neither study was at low risk of detection bias (Foley 2003; Krauß 2014).

Sustainability of effects

No studies reported physical function at three to six months after treatment or more than six months after treatment.

Subgroup analysis: type of exercise

One trial (105 participants) evaluated mixed exercise (Krauß 2014); and one trial (18 participants) evaluated resistance training, neuromuscular exercise, or both (Foley 2003). There was no evidence of a difference in effect according to type of exercise (test for subgroup differences: $P = 0.88$, $I^2 = 0\%$; Analysis 5.2).

Quality of life

No studies reported quality of life with placebo or attention control as the comparator.

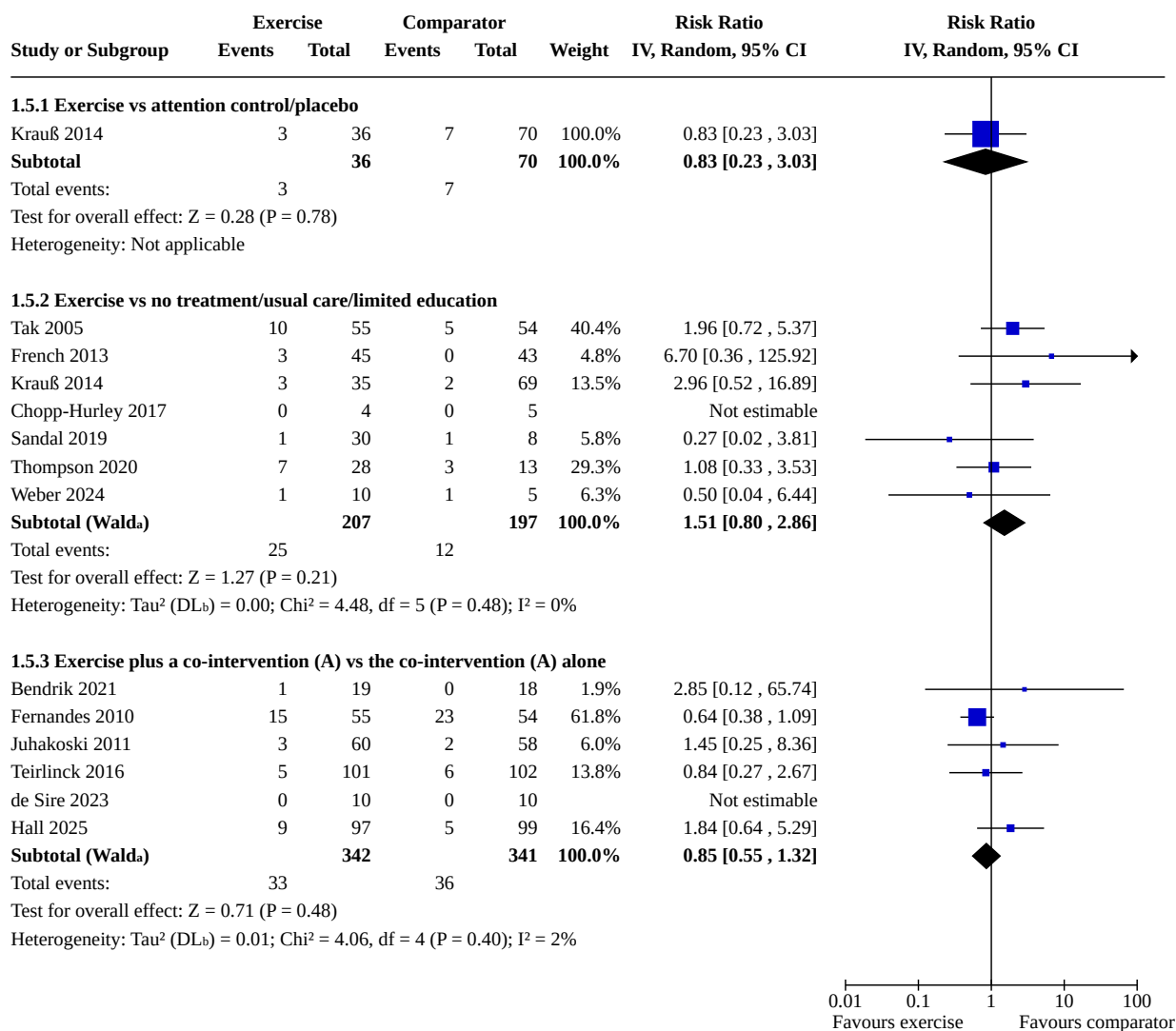
Participant-reported treatment success

No studies reported participant-reported treatment success with attention control or placebo as the comparator.

Study withdrawals

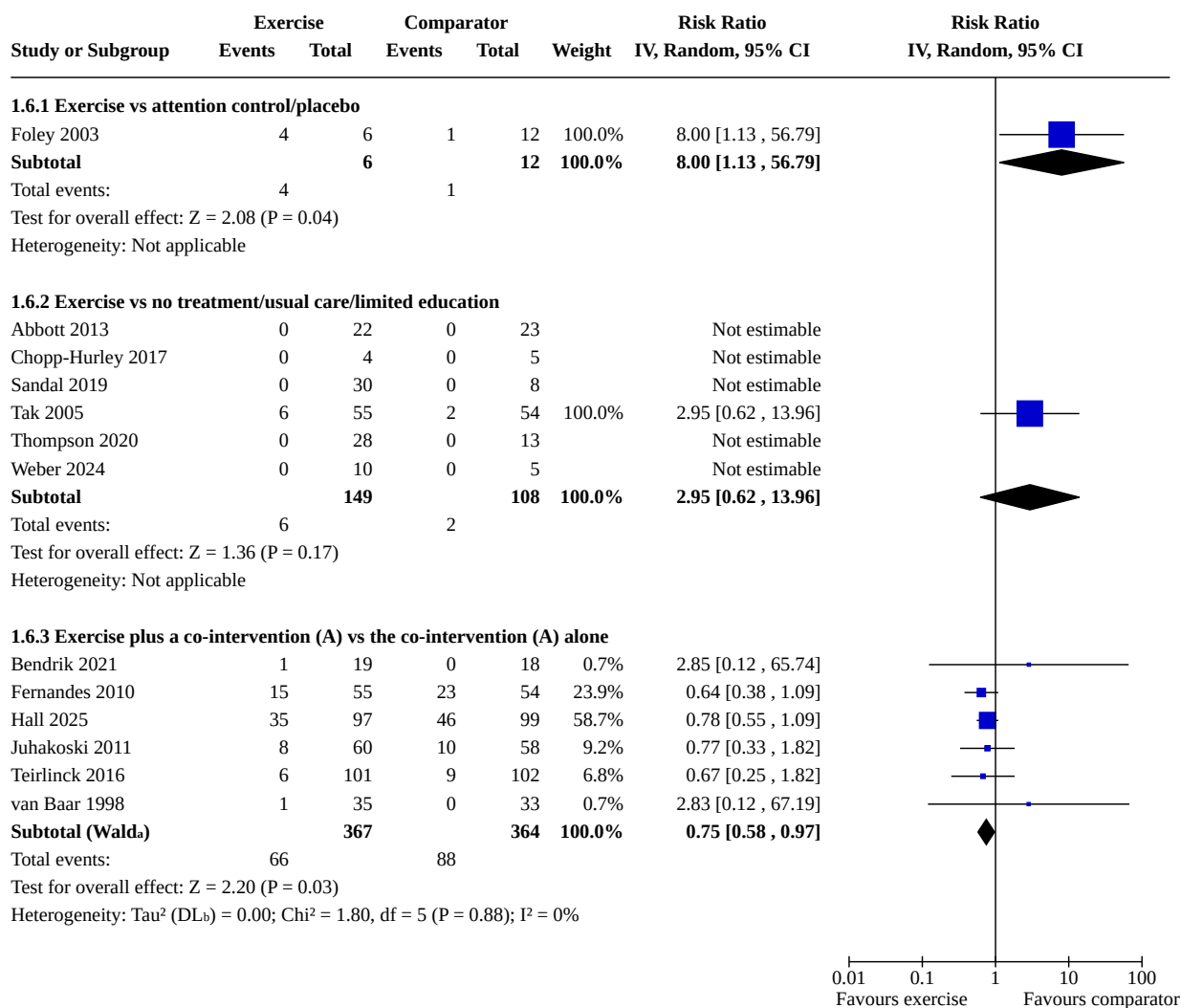
Exercise may have little to no effect on study withdrawals compared to placebo/attention control (RR 0.83, 95% CI 0.23 to 3.03; 1 study, 106 participants; Figure 5; Analysis 1.5; low-certainty evidence). We downgraded the evidence certainty one level for high risk of bias (included trial was at high risk of performance and detection bias) and one level for imprecision (low event rate resulted in a wide CI).

Figure 5. Effect on study withdrawals at final follow-up.

**Footnotes**^aCI calculated by Wald-type method.^b Tau^2 calculated by DerSimonian and Laird method.**Adverse events**

Exercise may increase the risk of adverse events compared to placebo/attention control, but the evidence is very uncertain (RR 8.00, 95% CI 1.13 to 56.79; 1 study, 18 participants; [Figure 6](#); Analysis

1.6; very low-certainty evidence). We downgraded the evidence certainty one level for high risk of bias (included study was at high risk of performance and detection bias) and two levels for imprecision (low event rate resulted in an extremely wide CI).

Figure 6. Effect on adverse events at final follow-up.**Footnotes**^aCI calculated by Wald-type method.^b Tau^2 calculated by DerSimonian and Laird method.**Exercise versus no treatment/usual care/limited education**

Ten studies (494 participants) assessed exercise compared with no treatment/usual care/limited education (Abbott 2013; Chopp-Hurley 2017; Fransen 2007; French 2013; Hopman-Rock 2000; Krauß 2014; Sandal 2019; Tak 2005; Thompson 2020; Weber 2024). [Summary of findings 2](#) presents the primary analysis results.

Pain

Exercise probably results in a slight reduction in pain immediately after treatment compared to no treatment/usual care/limited education (SMD -0.41, 95% CI -0.61 to -0.21; $I^2 = 3\%$; 9 studies, 449 participants; [Figure 3](#); Analysis 1.1; moderate-certainty evidence). These data represent a mean improvement of 7.19 points with exercise (MD -7.19 points, 95% CI -10.70 to -3.68). The CI of the MD did not meet the threshold for a clinically important difference (12 points). We downgraded the evidence certainty one level for

high risk of bias (> 50% of included studies were at high risk of performance and detection bias).

Sensitivity analyses

Our sensitivity analyses produced the following results ([Table 5](#)).

- Removal of one outlier with extreme results (15 participants; Weber 2024): SMD -0.38, 95% CI -0.58 to -0.18; equivalent to MD -6.66 points, 95% CI -10.17 to -3.16 ([Supplementary material 9](#)).
- Removal of two studies that used change scores (61 participants; Fransen 2007; Thompson 2020): SMD -0.41, 95% CI -0.63 to -0.20; equivalent to MD -7.19 points, 95% CI -11.05 to -3.51.
- Analyses of five studies (260 participants) at low risk of selection bias (Chopp-Hurley 2017; Fransen 2007; French 2013; Krauß 2014; Sandal 2019): SMD -0.49, 95% CI -0.75 to -0.22; equivalent to MD -8.59 points, 95% CI -13.15 to -3.86 (Analysis 4.1).
- Analysis of six studies (301 participants) at low risk of attrition bias (Chopp-Hurley 2017; Fransen 2007; French 2013; Krauß

2014; Thompson 2020; Sandal 2019): SMD -0.44 , 95% CI -0.68 to -0.20 ; equivalent to MD -7.72 points, 95% CI -11.92 to -3.51 (Analysis 4.2).

As no studies were at low risk of detection bias, we were unable to perform a sensitivity analysis based on this factor.

Sustainability of effects

Tak 2005 (109 participants) reported pain at three to six months after treatment (SMD -0.72 , 95% CI -1.11 to -0.33 ; equivalent to MD -12.62 points, 95% CI -19.46 to -5.79 ; Analysis 2.1). Abbott 2013 (45 participants) reported pain more than six months after treatment (SMD -0.29 , 95% CI -0.88 to 0.30 ; equivalent to MD -5.08 points, 95% CI -15.43 to 5.26 ; Analysis 3.1).

Subgroup analysis: type of exercise

Two studies (29 participants) evaluated mind-body exercise (Chopp-Hurley 2017; Fransen 2007); five studies (273 participants) evaluated mixed exercise (French 2013; Hopman-Rock 2000; Krauß 2014; Thompson 2020; Weber 2024); and two studies (147 participants) evaluated resistance training, neuromuscular exercise, or both (Sandal 2019; Tak 2005). There was no evidence of a difference in effect according to type of exercise (test for subgroup differences: $P = 0.27$, $I^2 = 24.7\%$; Analysis 5.3).

Physical function

Exercise probably results in a slight improvement in physical function immediately after treatment compared to no treatment/usual care/limited education (SMD -0.52 , 95% CI -0.71 to -0.32 ; $I^2 = 0\%$; 9 studies, 447 participants; Figure 4; Analysis 1.2; moderate-certainty evidence). These data represent a mean improvement of 8.79 points with exercise (MD -8.79 points, 95% CI -12.00 to -5.41). The CI of the MD did not meet the threshold for a clinically important difference (13 points). We downgraded the evidence certainty one level for high risk of bias ($> 50\%$ of included studies were at high risk of performance and detection bias).

Sensitivity analyses

Our sensitivity analyses produced the following results (Table 5).

- Removal of two studies using change scores (61 participants; Fransen 2007; Thompson 2020): SMD -0.48 , 95% CI -0.69 to -0.27 , equivalent to MD -8.11 points, 95% CI -11.66 to -4.56 .

- Analysis of five studies (260 participants) at low risk of selection bias (Chopp-Hurley 2017; Fransen 2007; French 2013; Krauß 2014; Sandal 2019): SMD -0.48 , 95% CI -0.74 to -0.21 ; equivalent to MD -8.11 points, 95% CI -12.50 to -3.55 (Analysis 4.4).
- Analysis of six studies (301 participants) at low risk of attrition bias (Chopp-Hurley 2017; Fransen 2007; French 2013; Krauß 2014; Thompson 2020; Sandal 2019): SMD -0.51 , 95% CI -0.75 to -0.26 ; equivalent to MD -8.62 points, 95% CI -12.67 to -4.39 (Analysis 4.5).

As no studies were at low risk of detection bias, we were unable to perform a sensitivity analysis based on this factor.

Sustainability of effects

Tak 2005 (109 participants) reported physical function at three to six months after treatment (SMD -0.44 , 95% CI -0.82 to -0.06 ; equivalent to MD -7.44 points, 95% CI -13.86 to -1.01 ; Analysis 2.2). Abbott 2013 (45 participants) reported physical function more than six months after treatment (SMD -0.38 , 95% CI -0.98 to 0.21 ; equivalent to MD -6.42 points, 95% CI -16.56 to 3.55 ; Analysis 3.2).

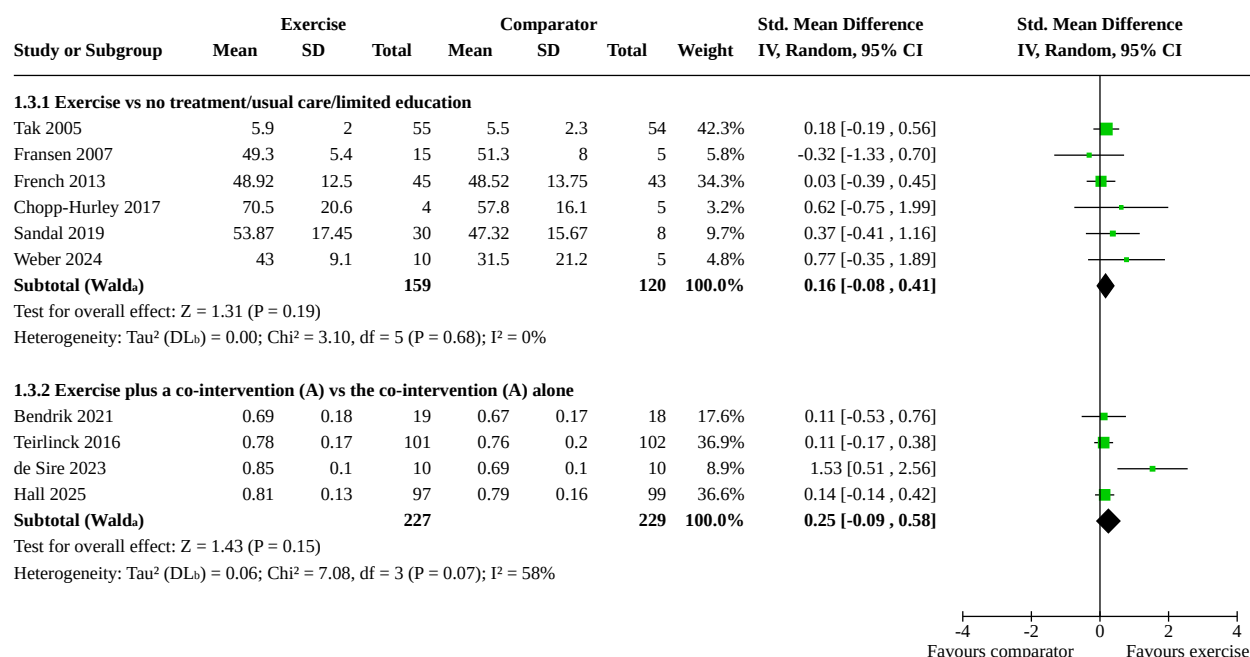
Subgroup analysis: type of exercise

Two studies (29 participants) evaluated mind-body exercise (Chopp-Hurley 2017; Fransen 2007), five studies (271 participants) evaluated mixed exercise (French 2013; Hopman-Rock 2000; Krauß 2014; Thompson 2020; Weber 2024), and two studies (147 participants) evaluated resistance training, neuromuscular exercise, or both (Sandal 2019; Tak 2005). There was no evidence of a difference in effect according to type of exercise (test for subgroup differences: $P = 0.63$, $I^2 = 0\%$; Analysis 5.4).

Quality of life

Exercise probably has little to no effect on quality of life immediately after treatment compared to no treatment/usual care/limited education (SMD 0.16 , 95% CI -0.08 to 0.41 ; $I^2 = 0\%$; 6 studies, 279 participants; Figure 7; Analysis 1.3; moderate-certainty evidence). These data represent a mean improvement of 2.31 points with exercise (MD 2.31 points, 95% CI -1.15 to 5.91). The CI of the MD did not meet the threshold for a clinically important difference (15 points). We downgraded the evidence certainty one level for high risk of bias ($> 50\%$ of included studies at high risk of performance and detection bias).

Figure 7. Effect on quality of life immediately after treatment.

**Footnotes**

aCI calculated by Wald-type method.

bTau² calculated by DerSimonian and Laird method.**Sensitivity analyses**

Our sensitivity analyses produced the following results (Table 5).

- Analysis of four studies (155 participants) at low risk of selection bias (Chopp-Hurley 2017; Fransen 2007; French 2013; Sandal 2019): SMD 0.09, 95% CI -0.24 to 0.43; equivalent to MD 1.30 points, 95% CI -3.46 to 6.20 (Analysis 4.7).
- Analysis of four studies (155 participants) at low risk of attrition bias (Chopp-Hurley 2017; Fransen 2007; French 2013; Sandal 2019): SMD 0.09, 95% CI -0.24 to 0.43; equivalent to MD 1.30 points, 95% CI -3.46 to 6.20 (Analysis 4.8).

As no studies were at low risk of detection bias, we were unable to perform a sensitivity analysis based on this factor.

Sustainability of effects

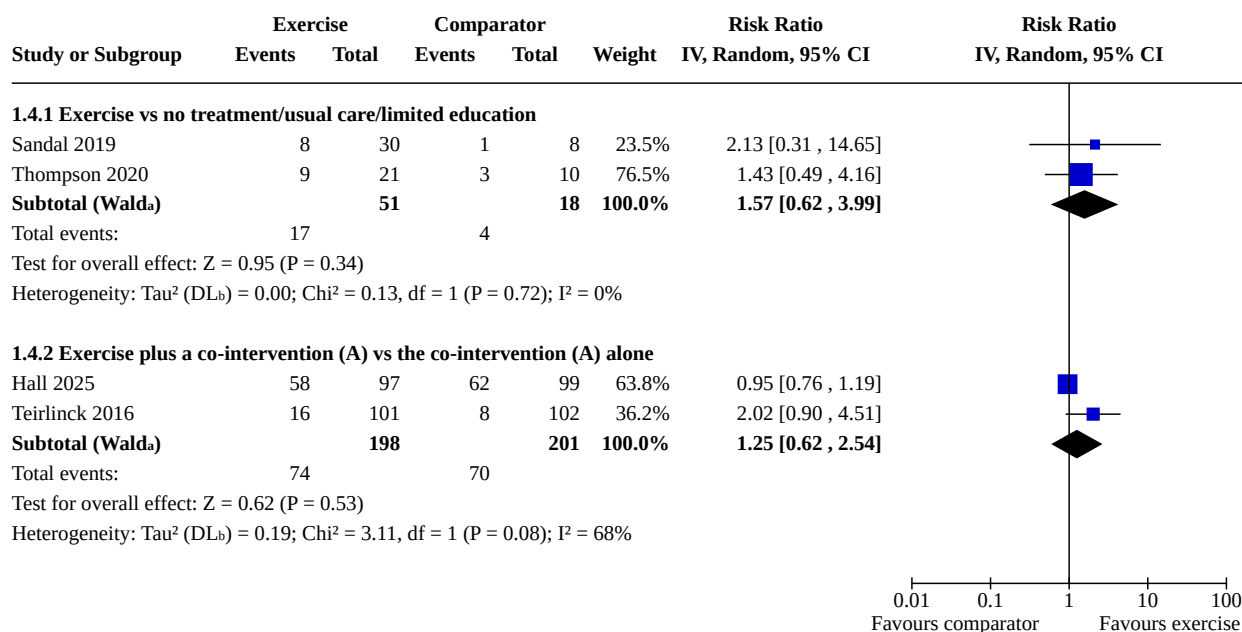
Tak 2005 (45 participants) reported quality of life at three to six months after treatment (SMD 0.54, 95% CI -0.05 to 1.14; equivalent to MD 7.79 points, 95% CI -0.72 to 16.44; Analysis 2.3).

Subgroup analysis: type of exercise

Two studies (29 participants) evaluated mind-body exercise (Chopp-Hurley 2017; Fransen 2007), two studies (103 participants) evaluated mixed exercise (French 2013; Weber 2024), and two studies (147 participants) evaluated resistance training, neuromuscular exercise, or both (Sandal 2019; Tak 2005). There was no evidence of a difference in effect according to type of exercise (test for subgroup differences: $P = 0.93$, $I^2 = 0\%$; Analysis 5.5).

Participant-reported treatment success

Exercise may have little to no effect on participant-reported treatment success compared to no treatment/usual care/limited education (RR 1.57, 95% CI 0.62 to 3.99; 2 studies, 69 participants; Figure 8; Analysis 1.4; low-certainty evidence). We downgraded the evidence certainty one level for high risk of bias (the included studies were at high risk of performance and detection bias), and one level for imprecision (low event rate resulted in a wide CI).

Figure 8. Effect on participant-reported treatment success immediately after treatment.**Footnotes**^aCI calculated by Wald-type method.^b Tau^2 calculated by DerSimonian and Laird method.**Study withdrawals**

Exercise may have little to no effect on study withdrawals compared to no treatment/usual care/limited education (RR 1.51, 95% CI 0.80 to 2.86; 7 studies, 404 participants; [Figure 5](#); Analysis 1.5; low-certainty evidence). We downgraded the evidence certainty one level for high risk of bias (> 50% of studies were at high risk of performance and detection bias) and one level for imprecision (low event rate resulted in a wide CI).

Adverse events

Exercise may have little to no effect on adverse events compared to no treatment/usual care/limited education, but the evidence is very uncertain (RR 2.95, 95% CI 0.62 to 13.96; 6 studies, 257 participants; [Figure 6](#); Analysis 1.6; very low-certainty evidence). We downgraded the evidence certainty one level for high risk of bias (> 50% of studies at high risk of performance and detection bias) and two levels for imprecision (low event rate resulted in an extremely wide CI).

Exercise plus a co-intervention (A) versus the co-intervention (A) alone

Seven studies assessed exercise plus a co-intervention (A) versus the co-intervention (A) alone (Bendrik 2021; de Sire 2023; Fernandes 2010; Hall 2025; Juhakoski 2011; Teirlinck 2016; van Baar 1998). [Summary of findings 3](#) presents the primary analysis results.

Pain

Exercise plus a co-intervention (A) probably has little to no effect on pain immediately after treatment compared to the co-intervention (A) alone (SMD -0.22, 95% CI -0.46 to 0.02; $I^2 = 57\%$; 7 studies, 751 participants; [Figure 3](#); Analysis 1.1; moderate-certainty evidence).

These data represent a mean improvement of 3.86 points with exercise (MD -3.86 points, 95% CI -8.07 to 0.35). The CI of the MD included no effect and did not meet the threshold for a clinically important difference (12 points). We downgraded the evidence certainty one level for high risk of bias (> 50% of included studies at high risk of performance and detection bias).

Sensitivity analyses

Our sensitivity analyses produced the following results ([Table 5](#)).

- Removal of one outlier with extreme results (20 participants; de Sire 2023): SMD -0.17, 95% CI -0.38 to 0.05; equivalent to MD -2.98 points, 95% CI -6.66 to 0.88 ([Supplementary material 9](#)).
- Analysis of five studies (694 participants) at low risk of selection bias (Fernandes 2010; Hall 2025; Juhakoski 2011; Teirlinck 2016; van Baar 1998): SMD -0.20, 95% CI -0.43 to 0.03; equivalent to MD -3.51 points, 95% CI -7.54 to 0.53 (Analysis 4.1).
- Analysis of two studies (216 participants) at low risk of detection bias (de Sire 2023; Hall 2025): SMD -0.56, 95% CI -1.48 to 0.37; equivalent to MD -9.82 points, 95% CI -25.95 to 6.49 (Analysis 4.3).

As all studies in the primary analysis were at low risk of attrition bias, we were unable to perform a sensitivity analysis based on that factor.

Sustainability of effects

Six studies (633 participants) reported pain at three to six months after treatment (Bendrik 2021; de Sire 2023; Fernandes 2010; Hall 2025; Teirlinck 2016; van Baar 1998; SMD -0.20, 95% CI -0.36 to -0.04; equivalent to MD -3.51 points, 95% CI -6.31 to -0.70; Analysis 2.1). Four studies (467 participants) reported pain more than six

months after treatment (Bendrik 2021; Fernandes 2010; Juhakoski 2011; Teirlinck 2016; SMD -0.24, 95% CI -0.42 to -0.06; equivalent to MD -4.21 points, 95% CI -7.36 to -1.05; Analysis 3.1).

Subgroup analysis: type of exercise

One study (196 participants) evaluated aerobic exercise (Hall 2025), and six studies (555 participants) evaluated mixed exercise (Bendrik 2021; de Sire 2023; Fernandes 2010; Juhakoski 2011; Teirlinck 2016; van Baar 1998). There was no evidence of a difference in effect according to exercise type (test for subgroup differences: $P = 0.79$, $I^2 = 0\%$; Analysis 5.6).

Physical function

Exercise plus a co-intervention (A) probably has little to no effect on physical function immediately after treatment compared to the co-intervention (A) alone (SMD -0.14, 95% CI -0.39 to 0.11; $I^2 = 61\%$; 7 studies, 751 participants; Figure 4; Analysis 1.2; moderate-certainty evidence). These data represent a 2.37-point improvement (MD -2.37 points, 95% CI -6.59 to 1.86). The CI of the MD did not meet the threshold for a clinically important difference (13 points). We downgraded the evidence certainty one level for high risk of bias (> 50% of included studies were at high risk of performance and detection bias).

Sensitivity analyses

Our sensitivity analyses produced the following results (Table 5).

- Removal of one outlier with extreme results (20 participants; de Sire 2023): SMD -0.08, 95% CI -0.28 to 0.12; equivalent to MD -1.35 points, 95% CI -4.73 to 2.03 (Supplementary material 9).
- Removal of one study using change scores (68 participants; van Baar 1998): SMD -0.20, 95% CI -0.45 to 0.04; equivalent to MD -3.38 points, 95% CI -7.60 to 0.68).
- Analysis of five studies (694 participants) at low risk of selection bias: SMD -0.08, 95% CI -0.31 to 0.15; equivalent to MD -1.35 points, 95% CI -5.24 to 2.53 (Analysis 4.4).
- Analysis of two studies (216 participants) at low risk of detection bias: SMD -0.68, 95% CI -2.05 to 0.70; equivalent to MD -11.49 points, 95% CI -36.64 to 11.83 (Analysis 4.6).

As all studies in the primary analysis were at low risk of attrition bias, we were unable to perform a sensitivity analysis based on that factor.

Sustainability of effects

Six studies (633 participants) reported physical function at three to six months after treatment (Bendrik 2021; de Sire 2023; Fernandes 2010; Hall 2025; Teirlinck 2016; van Baar 1998; SMD -0.24, 95% CI -0.46 to -0.01; equivalent to MD -4.06 points, 95% CI -7.77 to -0.17; Analysis 2.2). Four studies (467 participants) reported physical function more than six months after treatment (Bendrik 2021; Fernandes 2010; Juhakoski 2011; Teirlinck 2016; SMD -0.29, 95% CI -0.47 to -0.11; equivalent to MD -4.90 points, 95% CI -7.94 to -1.86; Analysis 3.2).

Subgroup analysis: type of exercise

One study (196 participants) evaluated aerobic exercise (Hall 2025), and six studies (555 participants) evaluated mixed exercise (Bendrik 2021; de Sire 2023; Fernandes 2010; Juhakoski 2011; Teirlinck 2016; van Baar 1998). There was no evidence of a difference in effect

according to exercise type (test for subgroup differences: $P = 0.62$, $I^2 = 0\%$; Analysis 5.7).

Quality of life

Exercise plus a co-intervention (A) probably has little to no effect on quality of life after treatment compared to the co-intervention (A) alone (SMD 0.25, 95% CI -0.09 to 0.58; Figure 7; Analysis 1.3; moderate-certainty evidence). These data represent a mean improvement of 3.60 points with exercise (MD 3.60 points, 95% CI -1.30 to 8.36). The CI of the MD did not meet the threshold for a clinically important difference (15 points). We downgraded the evidence certainty one level for high risk of bias (> 50% of included studies were at high risk of performance and detection bias).

Sensitivity analyses

Our sensitivity analyses produced the following results (Table 5).

- Analysis of two studies (399 participants) at low risk of selection bias: SMD 0.12, 95% CI -0.07 to 0.32; equivalent to MD 1.73, 95% CI -1.01 to 4.61 (Analysis 4.7).
- Analysis of two studies (216 participants) at low risk of detection bias: SMD 0.74, 95% CI -0.61 to 2.10; equivalent to MD 10.67 points, 95% CI -8.79 to 30.28 (Analysis 4.9).

As all studies in the primary analysis were at low risk of attrition bias, we were unable to perform a sensitivity analysis based on that factor.

Sustainability of effects

Three studies (419 participants) reported quality of life at three to six months after treatment (de Sire 2023; Hall 2025; Teirlinck 2016; SMD 0.15, 95% CI -0.05 to 0.34; equivalent to MD 2.16 points, 95% CI -0.72 to 4.90; Analysis 2.3). Two studies (260 participants) reported quality of life more than six months after treatment (Bendrik 2021; Teirlinck 2016; SMD 0.19, 95% CI -0.33 to 0.71; equivalent to MD 2.74 points, 95% CI -4.76 to 10.24; Analysis 3.3).

Subgroup analysis: type of exercise

One study (196 participants) evaluated aerobic exercise (Hall 2025), and three studies (260 participants) evaluated mixed exercise (Bendrik 2021; de Sire 2023; Teirlinck 2016). There was no evidence of a difference in effect according to exercise type (test for subgroup differences: $P = 0.50$, $I^2 = 0\%$; Analysis 5.8).

Participant-reported treatment success

Exercise plus a co-intervention (A) may have little to no effect on participant-reported treatment success compared to the co-intervention (A) alone (RR 1.25, 95% CI 0.62 to 2.54; 2 studies, 399 participants; Figure 8; Analysis 1.4; low-certainty evidence). We downgraded the evidence certainty one level for high risk of bias (the included studies were at high risk of performance and detection bias) and two levels for imprecision (low event rate resulted in an extremely wide CI).

Study withdrawals

Exercise plus a co-intervention (A) probably has little to no effect on study withdrawals compared to the co-intervention (A) alone (RR 0.85, 95% CI 0.55 to 1.32; 6 studies, 683 participants; Figure 5; Analysis 1.5; moderate-certainty evidence). We downgraded the

evidence certainty one level for high risk of bias (> 50% of studies at high risk of performance and detection bias).

Adverse events

Exercise plus a co-intervention (A) probably reduces the risk of adverse events slightly compared to the co-intervention (A) alone (RR 0.75, 95% CI 0.58 to 0.97; 6 studies, 731 participants; [Figure 6](#); Analysis 1.6; moderate-certainty evidence). We downgraded the evidence certainty one level for high risk of bias (> 50% of studies at high risk of performance and detection bias).

DISCUSSION

Summary of main results

Eight eligible RCTs have been published since the prior version of this Cochrane review [17]. In total, this update included 18 studies (1368 participants). Two studies compared exercise to attention control or placebo, 10 compared exercise to no treatment/usual care/limited education, and seven evaluated exercise plus a co-intervention (A) versus the co-intervention (A) alone. One study included two comparator categories. Exercise interventions ranged from two to 52 weeks in duration.

Compared to attention control or placebo, exercise may have little to no effect on pain and may slightly improve physical function immediately after treatment ([Summary of findings 1](#)). Based on the MCID we used, the benefit for physical function was of uncertain clinical importance. It is unknown whether the treatment effects were sustained, as both studies in this comparison only measured the outcomes immediately after treatment. Neither study assessed quality of life or participant-reported treatment success. Exercise compared to attention control or placebo may have little to no effect on study withdrawals and may increase the risk of adverse events, although the evidence for adverse events is very uncertain.

Compared to no treatment, usual care, or limited education, exercise probably improves pain and physical function slightly ([Summary of findings 2](#)). However, based on the MCIDs we used [33], as well as other established MCIDs [110], the benefits were of uncertain clinical importance. Evidence from single trials suggested that benefits for pain and physical function were sustained at three to six months, but not in the longer term. Exercise probably has little to no effect on quality of life immediately after treatment. We found that exercise may have little to no effect on participant-reported treatment success immediately after treatment, or on study withdrawals or adverse events, but the evidence was very uncertain for adverse events.

Exercise plus a co-intervention (A), compared to the co-intervention (A) alone, probably has little to no effect on pain, physical function, or quality of life immediately after treatment ([Summary of findings 3](#)). Exercise may improve pain and physical function slightly in the longer term (at 3–6 months and > 6 months). We found that exercise may have little to no effect on participant-reported success immediately after treatment, probably has little to no effect on study withdrawals, and probably reduces the risk of adverse events slightly.

Subgroup analyses produced no evidence of any differences between types of exercise across the comparisons.

Limitations of the evidence included in the review

In all comparators, we downgraded the evidence certainty for all outcomes due to risk of bias, as more than half of the studies were at high risk of performance and detection bias. Blinding of participants and personnel is challenging in exercise trials. Nevertheless, failure to blind may induce bias in effect estimations, resulting in overestimation of the benefits of exercise [111]. Improvements with exercise would probably have been even smaller if studies had addressed these biases. We downgraded for imprecision in all comparisons, as low event rates limited our confidence that the true effect was close to the estimate. Of the 18 included studies, 15 contributed final values and three contributed change-from-baseline data. It is usually advised to not pool these measures in a meta-analysis of SMDs, because the SDs used in the standardisation reflect different sources of variability. However, sensitivity analyses excluding the three change-score studies did not meaningfully change the results.

Overall, the studies included participants whose findings are largely generalisable to people with hip osteoarthritis across different countries and settings assessed in this review. Studies were conducted across 12 high-income countries, most frequently the Netherlands and Australia. Participants were mostly women (63%), with mean ages ranging from 53 years to 74 years, and mean BMI ranging from 24.7 kg/m² to 30.2 kg/m². Because there is a relatively poor association between radiographic structural changes and symptoms in hip osteoarthritis [8], we chose not to extract data relating to structural severity of osteoarthritis among study participants. The range of samples in our review, with varying symptoms at baseline, suggest that our findings are broadly generalisable. However, applicability to younger adults with hip osteoarthritis may be limited, as they were under-represented in the included studies.

The exercise programmes included in our review varied substantially in terms of type, frequency, and duration. Interventions ranged from two to 52 weeks in duration. Some exercise interventions included components of behaviour change strategies to promote adherence, but inconsistently, which may have contributed to heterogeneity in results. The observed heterogeneity may influence the pooled estimates by widening CIs and reducing certainty in the precise magnitude of effects. This heterogeneity in exercise interventions likely explains the moderately high I² statistics we observed in some of our meta-analyses. Removing outliers reduced the heterogeneity somewhat.

Most studies assessed the effects of exercise on pain and physical function immediately after treatment. Fewer trials assessed quality of life (56%), participant-reported treatment success (22%), and adverse events (72%). As such, we are less certain of the applicability of results for these outcomes. Several studies had small sample sizes (56% of studies analysed fewer than 50 participants). Additional small trials of exercise for hip osteoarthritis may further inflate effect estimates and uncertainty in the evidence. Only half of studies reported longer-term outcomes, measured from three months after the intervention. Therefore, we are less certain about the effects of exercise after the exercise programme stops. We analysed the latest follow-up (after six months) to capture the longest-term effect of exercise. Follow-up duration varied across studies (12–29 months). We analysed the latest available follow-up for each study to reflect the sustained effects of exercise rather than shorter-term effects. This

variation in duration may contribute to heterogeneity and should be considered when interpreting the effects. However, the direction of effect was broadly similar across shorter and longer follow-up periods.

Limitations of the review processes

When conducting this review, we adhered to standard Cochrane methods. To minimise bias, up to eight review authors independently screened and selected studies. Two review authors independently rated risk of bias to ensure consistency in judgements, and an independent methodological expert undertook GRADE assessments for all outcomes. Although a Cochrane Information Specialist conducted a comprehensive search of published and ongoing studies across multiple databases, we may still have missed some relevant studies. We were unable to obtain data from four eligible trials, and nine ongoing trial records may also meet inclusion criteria once completed. If a substantial proportion of these unavailable or ongoing trials are of high quality (e.g. low risk of bias and adequately powered), their inclusion could shift the pooled treatment effects towards the null. In addition, the review methods did not enable reliable detection of rare or serious adverse events, as these outcomes were infrequently reported, and sample sizes were small. Consequently, conclusions regarding safety should be interpreted with caution.

Agreements and disagreements with other studies or reviews

The prior version of this Cochrane review, which included 10 trials, concluded that there was high-quality evidence that exercise slightly improves pain and physical function immediately after treatment [17]. In this update, we analysed results according to comparator group, whereas the previous review pooled all trials regardless of comparator. Our findings for exercise compared with no treatment/usual care/limited education are largely consistent with the earlier review. The effect sizes are in line with those observed in a 2023 systematic review and meta-analysis on exercise for pain and function in hip osteoarthritis [112, 113]. Our GRADE ratings for pain and physical function were lower than those of the previous review update, reflecting the high overall risk of performance and detection bias among included trials, as well as imprecision in some estimates. The clinical importance of the small improvements is questionable, given how far the pooled estimates fall below commonly used MCIDs.

For quality of life, we found moderate-certainty evidence that exercise has little to no effect compared with no treatment/usual care/limited education, or when added to another intervention and compared with the other intervention alone. This conclusion is consistent with the previous review, increasing our certainty in the results [17]. These findings also align with another systematic review examining the impact of exercise on quality of life [114]. Only three trials assessed participant-reported treatment success, and the evidence suggested little to no improvement in this outcome with exercise. As reported in the prior review update [17], exercise may not increase adverse events, although our certainty is lower due to sparse event data. Exercise is unlikely to increase study withdrawals, again consistent with earlier findings. To our knowledge, no systematic reviews since the previous Cochrane review have specifically evaluated adverse events associated with exercise for hip osteoarthritis [17].

We found that effects on pain, physical function, and quality of life were similar for different types of exercise. This finding contrasts with a previous network meta-analysis that found that mixed exercise programmes were less effective than aerobic and mind-body programmes for people with hip/knee osteoarthritis [114]. However, generalising findings from knee osteoarthritis to hip osteoarthritis is problematic given potential differences between these conditions [115]. Several high-quality trials have directly compared different exercise types for hip osteoarthritis [116, 117], but they were ineligible for inclusion in this review because they lacked an eligible control group. This limits the strength and generalisability of our subgroup findings.

This review update expands on the previous version by assessing the effects of exercise according to type of comparator group. Other systematic reviews on the effects of exercise on hip/knee osteoarthritis have found differential effects based on comparator types [118]. While we found no statistical evidence that the effects of exercise differed according to comparator type, the interpretation of the effect estimates varied across comparators.

AUTHORS' CONCLUSIONS

Implications for practice

We found very low- to moderate-certainty evidence that exercise may improve pain and physical function slightly, but have little to no effect on quality of life, in the short-term. Based on the minimal clinically important difference (MCID) thresholds we applied, potential benefits were not clinically meaningful. However, interpreting clinical relevance with MCIDs requires caution. The thresholds we used were derived largely from knee and mixed osteoarthritis populations, which may not reflect hip osteoarthritis. They were also established for the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) instrument [33], yet our pooled analyses included studies that had used other pain and function instruments. Finally, MCIDs are context dependent and vary by baseline symptom severity, follow-up timing, intervention characteristics, and participant characteristics [119, 120]. Therefore, MCIDs should be considered a supplementary guide alongside factors such as cost, convenience, and safety of exercise interventions [121, 122].

Adverse events were poorly reported. We found that adverse events might increase with exercise compared to attention control/placebo, but the evidence is very uncertain. Thus, based on the evidence available, and considering the accessibility and broader benefits of this treatment to combat other comorbidities associated with hip osteoarthritis (e.g. metabolic syndrome [5]), exercise is likely advisable in this population.

Exercise is often prescribed as part of a broader physiotherapy programme in clinical practice, and our findings are only applicable to the effect of exercise alone. Thus, caution is necessary when extrapolating to combined clinical programmes.

Implications for research

We found limited studies specific to hip osteoarthritis (50%), many of which had small sample sizes. Smaller studies are known to overestimate treatment effects [123], making a true large benefit unlikely. Nevertheless, adequately powered studies are needed to provide precise and reliable estimates of the true effect. There is also a need for diligent selection of comparison groups to

best contribute to evidence about the effect of exercise on hip osteoarthritis management. We acknowledge the methodological challenges of designing fair and appropriate comparators. To support future trial design, we recommend that trialists consult the comprehensive guidance on selecting appropriate control groups for both mechanistic and pragmatic exercise studies [124, 125, 126]. Future research should also assess the recommended core outcome set, including health-related quality of life and perceived effect [127]. We found these important outcomes, particularly perceived effect, were available for a subset of studies only. Studies should also include the systematic measurement of harms. Adverse event reporting was inconsistent across studies, with many studies providing limited detail or unclear definitions. To improve the certainty of safety assessments, future studies should incorporate standardised and proactive adverse event monitoring protocols. Consistent and comprehensive safety monitoring will enable more reliable estimation of harms associated with exercise for hip osteoarthritis.

Most included studies were at high risk of performance and detection bias. Strategies to minimise performance and detection bias include the use of attention-control comparators and incorporating objective outcomes (e.g. functional measures) that can be assessed in a blinded manner. Greater consumer involvement in trial design, particularly in selecting outcomes that reflect patient priorities and in developing strategies to improve adherence, may further enhance the relevance and applicability of future evidence. Finally, research is needed to establish hip-specific MCIDs for exercise to improve clinical interpretation. Researchers could consider complementary metrics such as the smallest worthwhile effects, which account for benefits, harms and inconveniences, and costs [128, 129].

We concur with the European Alliance of Associations for Rheumatology and advocate for well-planned and conducted research to understand the mechanisms of exercise effects on osteoarthritis symptoms [16]. Until the mechanisms underlying exercise's effects on symptoms are better understood, clinicians and patients may continue to see improvements that fall short of clinical relevance.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD007912.pub3](https://doi.org/10.1002/14651858.CD007912.pub3).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Analyses

Supplementary material 7 Data package

Supplementary material 8 Supplementary Table 1. Baseline demographic and clinical characteristics for the trial participants

Supplementary material 9 Sensitivity analyses

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

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The following people conducted the editorial process for this article.

- Sign-off Editor (final editorial decision): Joanne V Glinsky, University of Sydney, Sydney, Australia.
- Managing Editor (provided editorial guidance to authors, edited the article): Ben Ridley, Cochrane Editorial Service.
- Editorial Assistant (selected peer reviewers, conducted editorial policy checks, collated peer-reviewer comments and supported editorial team): Andrew Savage, Cochrane Editorial Service.
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Contributions of authors

Conceptualisation: all review authors.

Investigation: all review authors independently screened studies and contributed to data extraction from eligible studies.

Formal analysis: MH.

Validation: MH and BJL. HPF assessed risk of bias for the trial authored by MH.

Methodology: all review authors.

Supervision: MH.

Writing – original draft: MH and BJL.

Writing – review & editing: all review authors.

Review authors were not involved in eligibility decisions, data extraction, risk of bias assessment, or certainty of evidence grading for studies of which they were authors.

Some review authors involved in previous versions of the review published in 2009 and 2014 are not included in the current author byline (Sara McConnell and Marlene Fransen). Some retained content from earlier versions reflects their contributions.

Declarations of interest

Michelle Hall

- Grant funds from Medical Research Future Fund
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- Editorial Board Member role for Arthritis Care and Research
- Osteoarthritis Research Society International Board Member who has travel expenses to meetings paid
- Publication of papers expressing opinions relevant to this work
- Co-author of included study (Hall 2025)

Belinda Lawford

- Steering Committee member of Osteoarthritis Research Society International Rehabilitation Group

Kim Bennell

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Rana Hinman

- Grant funds from National Health and Medical Research Council
- Publication of papers and social media expressing opinion relevant to this work
- Editorial Board Member role for Journal of Physiotherapy
- Research funding from Medibank for projects related to exercise and osteoarthritis
- Co-author of included study (Hall 2025)

Libby Spiers

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Helen French

- Co-author of included study (French 2013)

Gabriela Hernandez-Molina

- No relevant interests

Alex Kimp

- No relevant interests

Stephan Reichenbach

- No relevant interests

Michelle Hall, Libby Spiers, Fiona Dobson, Rana Hinman, Helen French, and Kim Bennell are authors of studies included in this review. None of these authors were involved in the assessment of study eligibility, data extraction, or assessment of risk of bias for their own studies. These authors did not perform the GRADE assessment.

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Registration and protocol

The original protocol was for a review on exercise for osteoarthritis of the hip or knee (<https://doi.org/10.1002/14651858.CD004376>). Two reviews of separate conditions (knee arthritis and hip arthritis) followed. The first review on hip osteoarthritis was published in 2009 (<https://doi.org/10.1002/14651858.CD007912>), and the first update was published in 2014 (<https://doi.org/10.1002/14651858.CD007912.pub2>).

Data, code and other materials

As part of the published Cochrane review, the data package is made available for download for users of the Cochrane Library ([Supplementary material 7](#)). The data package includes study-level data (e.g. study characteristics, study arms, references, and outcome data), consensus risk of bias assessments, and analysis datasets (including overall estimates and individual study data used in meta-analyses). Where applicable, subgroup estimates and additional reference lists (e.g. studies awaiting classification) are also included. Appropriate permissions have been obtained for all shared materials. Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods. No additional scripts or external analytical tools were used. Data extraction was conducted using Covidence, and data extraction forms are available on reasonable request.

What's new

Date	Event	Description
23 July 2026	New search has been performed	New search run 5 February 2025; 3885 new records identified; eight new studies added to this update.
23 July 2026	New citation required and conclusions have changed	Authors changed; methods updated; 8 new studies included; certainty of evidence downgraded (due to high number of trials at

Date	Event	Description
		unclear or high risk of bias, some publication bias and imprecision); conclusion updated ("We found very low to low certainty evidence that exercise may result in a slight improvement in pain and function, but not quality of life in the short-term. However, based on the thresholds for minimal important differences used, the benefits were of questionable clinical importance.")

History

Review first published: Issue 3, 2009

Date	Event	Description
24 March 2014	New citation required and conclusions have changed	Change in conclusions on update: significant benefit in terms of physical function now demonstrated. Methods were updated in accordance with current Cochrane Collaboration recommendations: risk of bias assessment and Summary of Findings Tables added. Quality of life assessment and study withdrawal rates were added in the update. Pain and physical function outcomes were further disaggregated into immediate post treatment effects and sustainability (3-6 months post treatment)
9 May 2013	New search has been performed	Five new studies added to this update: Fernandes 2010; Juhakoski 2011; Carlson 2011; French 2013; Abbott 2013.
19 April 2009	New citation required but conclusions have not changed	Substantive amendment
14 January 2009	Amended	Converted to new review format. CMSG ID added A040-R
14 January 2009	New search has been performed	This updated review is one of two Cochrane reviews replacing an earlier review, 'Exercise for osteoarthritis of the hip or knee'. Since the original review, the editors decided to subdivide the review into separate conditions. The Background section has been revised to provide information on the specific disorder only, and the search strategy has been revised accordingly. The Methods section has been updated to reflect current Cochrane Musculoskeletal Group methods. 3 new studies were added in this updated review: Foley 2003; Fransen 2007; Tak 2005

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ADDITIONAL TABLES

Table 1. Overview of synthesis and included studies

Study ID (country)	Study design	Treatment groups	Sample size	Outcome domains	Specific outcome measure	Follow-up from randomisation (relative to treatment)
Abbott 2013 (New Zealand)	RCT	Exercise	22	1. Pain	1. WOMAC pain	12 months (> 6 months)
		Usual care	23	2. Physical function 3. Adverse events ^a	2. WOMAC function subscale	
Bendrik 2021 (Sweden)	RCT	Exercise + education	19	1. Pain	1. HOOS pain subscale	6 months (immediately after treatment), 12 months (3–6 months) and 24 months (> 6 months)
		Education	18	2. Physical function 3. Quality of life 4. Study withdrawals ^a 5. Adverse events ^a	2. HOOS function subscale 3. EQ-5D	
Chopp-Hurley 2017 (Canada)	RCT	Exercise	4	1. Pain	1. HOOS pain subscale	12 weeks (immediately after treatment)
		No exercise	5	2. Physical function 3. Quality of life 4. Study withdrawals ^a 5. Adverse events ^a	2. HOOS function subscale 3. HOOS quality of life subscale	
de Sire 2023 (Italy)	RCT	Proprioceptive neuromuscular facilitation + conventional rehabilitation	10	1. Pain 2. Physical function 3. Study withdrawals ^a	1. Global/overall pain NRS 2. Harris Hip Score	2 weeks (immediately after treatment), 6 months (3–6 months)
		Conventional rehabilitation	10			
Fernandes 2010 (Norway)	RCT	Exercise + education	55	1. Pain	1. Activity pain NRS	4 months (immediately after treatment), 10 months (3–6 months), 29 months (> 6 months)
		Education	54	2. Physical function 3. Study withdrawals ^a 4. Adverse events ^a	2. WOMAC function subscale	
Foley 2003 (Australia)	RCT	Exercise	6	1. Pain	1. WOMAC pain subscale	6 weeks (immediately after treatment)
		No exercise	12	2. Physical function 3. Adverse events ^a	2. WOMAC function subscale	
Fransen 2007 (Australia)	RCT	Exercise	15	1. Pain	1. WOMAC pain	12 weeks (immediately after treatment)
		Waiting list	5	2 Physical function	2. WOMAC function subscale	

Table 1. Overview of synthesis and included studies (Continued)

				3. Quality of life	3. SF-12 MCS	
French 2013 (Ireland)	RCT	Exercise	45	1. Pain	1. Activity pain NRS	9 weeks (immediately after treatment)
		Waiting list	43	2. Physical function	2. WOMAC function subscale	
				3. Quality of life	3. SF-36 MCS	
				4. Study withdrawals ^a		
Hall 2025 (Australia)	RCT	Resistance and aerobic exercise	97	1. Pain	1. Global/overall pain NRS	3 months (immediately after treatment), 9 months (6 months after treatment)
		Resistance exercise	99	2. Physical function	2. WOMAC function subscale	
				3. Quality of life	3. AQoL-6D	
				4. Participant-reported success ^b	4. 7-point Likert scale (from "much worse" to "much better")	
				5. Study withdrawals ^a		
				6. Adverse events ^a		
Hopman-Rock 2000 (Netherlands)	RCT	Exercise	11	1. Pain	1. Global/overall VAS	6 weeks (immediately after treatment)
		No exercise	13	2. Physical function	2. IRGL function	
Juhakoski 2011 (Finland)	RCT	Exercise + GP care	60	1. Pain	1. WOMAC pain subscale	3 months (immediately after treatment), 24 months (> 6 months)
		GP care	58	2. Physical function	2. WOMAC function subscale	
				3. Study withdrawals ^a		
				4. Adverse events ^a		
Krauß 2014 (Germany)	RCT	Exercise	71	1. Pain	1. WOMAC pain subscale	3 months (immediately after treatment)
		Usual care	69	2. Physical function	2. WOMAC function subscale	
		Placebo	70	3. Study withdrawals ^a		
Sandal 2019 (Denmark)	RCT	Exercise	30	1. Pain	1. HOOS pain scale	8 weeks (immediately after treatment)
		Waiting list	8	2. Physical function	2. HOOS function subscale	
				3. Quality of life	3. HOOS quality of life subscale	
				4. Participant-reported success ^b	4. 7-point Likert scale (from "somewhat worse" to "markedly better")	
				5. Study withdrawals ^a		
				6. Adverse events ^a		
Tak 2005 (Netherlands)	RCT	Exercise	55	1. Pain	1. Global/overall pain VAS	8 weeks (immediately after treatment), 5 months (3–6 months)
		No exercise	54	2. Physical function	2. GARS function	
				3. Quality of life	3. Generic quality of life	
				4. Study withdrawals ^a		

Table 1. Overview of synthesis and included studies (Continued)

5. Adverse events ^a						
Teirlinck 2016 (Netherlands)	RCT	Exercise and GP care	101	1. Pain	1. Global/overall pain NRS	3 months (immediately after treatment), 12 months (> 6 months)
		GP care	102	2. Physical function	2. HOOS function subscale	
				3. Quality of life	3. EQ-5D	
				4. Participant-reported success ^b	4. 7-point Likert scale (from "total recovery" to "worse than ever")	
				5. Study withdrawals ^a		
				6. Adverse events ^a		
Thompson 2020 (USA)	RCT	Exercise	28	1. Pain	1. Global/overall VAS	3 months (immediately after treatment)
		Waiting list	13	2. Physical function	2. WOMAC function subscale	
				3. Participant-reported success ^b	3. "Responders" defined as participants who achieved a minimal clinically important response (≥ 13 mm on pain VAS)	
				4. Study withdrawals ^a		
				5. Adverse events ^a		
van Baar 1998 (Netherlands)	RCT	Exercise and GP care	35	1. Pain	1. Global/overall pain VAS	3 months (immediately after treatment), 9 months (3–6 months)
		GP care	33	2. Physical function	2. IRGL function	
				3. Adverse events ^a		
Weber 2024 (Germany)	RCT	Exercise	10	1. Pain	1. HOOS pain subscale	3 months (immediately after treatment)
		Usual care	5	2. Physical function	2. HOOS function subscale	
				3. Quality of life	3. HOOS quality of life subscale	
				4. Study withdrawals ^a		
				5. Adverse events ^a		

^a Measure taken at final follow-up.

^b Measure taken immediately after treatment.

AQoL-6D: Six-dimension Assessment of Quality of Life; **EQ-5D:** EuroQol Five-Dimension Questionnaire; **GARS:** Groningen Activity Restriction Scale; **GP:** general practitioner; **HOOS:** Hip Disability and Osteoarthritis Outcome Score; **IRGL:** Influence of Rheumatic Disease on General Health and Lifestyle; **NRS:** numeric rating scale; **RCT:** randomised controlled trial; **SF-12 MCS:** Medical Outcomes Study 12-Item Short-Form Health Survey Mental Component Summary; **SF-36 MCS:** Medical Outcomes Study 36-Item Short-Form Health Survey Mental Component Summary; **VAS:** visual analogue scale; **WOMAC:** Western Ontario and McMaster Universities Osteoarthritis Index.

Table 2. Characteristics of interventions of included trials

Study ID	Comparison	Description of exercise intervention	Description of control
Abbott 2013	Exercise vs usual care	Mix of clinic/home, mix of supervised/unsupervised individual setting, delivered entirely in person Aerobic + resistance + neuromuscular	All participants continued to receive routine care offered by their own GP

Table 2. Characteristics of interventions of included trials (Continued)

		156 sessions (3 × per week, 52 weeks)	and other health providers. No trial interventions were provided.
Bendrik 2021	Exercise + education vs education	3 consultations with physiotherapist, delivered entirely in person Individualised programme addressing type of exercise, form, frequency, and dose Aerobic + resistance	Education
Chopp-Hurley 2017	Exercise vs no intervention	Clinic, supervised group setting, delivered entirely in person Yoga 36 sessions (3 × per week; 60 minutes, over 12 weeks)	No intervention
de Sire 2023	Manual therapy + exercise vs manual therapy	Clinic, supervised individual setting, delivered entirely in person Stretching (proprioceptive neuromuscular facilitation) 10 sessions (5 × per week; 45 minutes, over 2 weeks)	Passive/active kinesiotherapy techniques (concentric and eccentric isotonic contractions), stretching techniques (passive in relaxation, whereby the therapist brings the muscular chain into tension and holds it for 60 seconds)
Foley 2003	Exercise vs attention control	Clinic, supervised group setting, delivered entirely in person Resistance + range of motion 18 sessions (3 × per week over 6 weeks)	Fortnightly telephone calls to check condition, drug use, or injuries
Fransen 2007	Exercise vs waiting list	Clinic, supervised group setting, delivered entirely in person Tai chi 24 sessions (2 × per week, 60 minutes for 12 weeks)	Waiting list
French 2013	Exercise vs waiting list	Mix of clinic/home, mix of supervised/unsupervised, individual setting, delivered entirely in person Resistance + flexibility 48–64 sessions (6–8 sessions lasting 30 minutes, over 8 weeks)	Waiting list
Hall 2025	Resistance and aerobic exercise vs resistance exercise	Home-based programme, consultations with physiotherapists, delivered online Resistance (3 sessions per week for 9 months) Aerobic (150 minutes per week at moderate intensity for 9 months) 9 consultations over 3 months	Home-based programme, consultations with physiotherapists, delivered online Resistance (3 sessions per week for 9 months)

Table 2. Characteristics of interventions of included trials (Continued)

			9 consultations over 3 months
Hopman-Rock 2000	Exercise vs waiting list	Mix of clinic/home, mix of supervised/unsupervised, mix of group/individual setting, delivery entirely in person Resistance + mixed aerobic 24 sessions (4 × per week, 60 minutes for 6 weeks)	Waiting list
Juhakoski 2011	Exercise + usual care vs usual care	Mix of clinic/home, mix of supervised/unsupervised, mixed group/individual setting, delivered entirely in person Resistance + stretching 12 supervised (once per week, 45 minutes duration) exercise sessions. 4 additional booster sessions 1 year later (between 12 and 13 months of follow-up) 3 times per week	Normal routine care offered by participants' own GP (analgesics and physiotherapy)
Krauß 2014	Exercise vs no intervention vs placebo	Mix of clinic/home, mix of supervised/unsupervised, mixed group/individual setting, delivered entirely in person Resistance + flexibility + balance + proprioception 12 supervised (once per week, 60–90 minutes), 24 unsupervised (twice per week, 30–40 minutes) over 12 weeks	No intervention and placebo
Sandal 2019	Exercise vs waiting list	Clinic, supervised group setting, delivered entirely in person Neuromuscular: strengthening + functional + core stability + postural orientation 16 sessions (twice per week); 60 minutes over 8 weeks	Waiting list
Tak 2005	Exercise vs no intervention	Clinic, supervised group setting, delivered entirely in person Mixed resistance + aerobic 8 sessions (once per week); 60 minutes over 8 weeks	No intervention except self-initiated contact with own GP
Teirlinck 2016	Exercise + usual care vs usual care	Clinic, supervised individual setting, delivered entirely in person Resistance + flexibility Up to 12 sessions; 30 minutes, over 12 weeks. 3 booster sessions in months 5, 7, 9.	Usual care given by participants' own GP and information about hip osteoarthritis
Thompson 2020	Exercise vs waiting list	Mix of clinic/home, mix of supervised/unsupervised, mixed group/individual setting, delivered entirely in person Resistance + flexibility + endurance 36 sessions lasting 60 minutes each (3 × per week for 12 weeks)	Waiting list
van Baar 1998	Exercise + education vs education	Mix clinic/home, mix supervised/unsupervised individual setting, delivery entirely in person Resistance 36 sessions (3 sessions per week over 12 weeks)	Education

Table 2. Characteristics of interventions of included trials (Continued)

Weber 2024	Exercise vs usual care	Digital mobile application, delivery entirely via mobile application Neuromuscular exercise + physical activity Neuromuscular exercise 3 days and physical activity 3 days; 12 weeks	Usual care provided by physiotherapist
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GP: general practitioner.

Table 3. Outcome reporting bias in trials (ORBIT) matrix

Study ID	Pain	Function	Quality of life	Participant-reported success	Adverse events
Abbott 2013	WOMAC pain subscale	WOMAC function subscale	No measure	No measure	Reported
Bendrik 2021	HOOS pain subscale	HOOS function subscale	EQ-5D	No measure	Reported
Chopp-Hurley 2017	HOOS pain subscale	HOOS function subscale	HOOS quality of life subscale	No measure	Not reported
de Sire 2023	Global/overall pain NRS	Harris Hip Score	EQ-5D	No measure	Not reported
Fernandes 2010	Activity pain NRS	WOMAC function subscale	No measure	No measure	Reported
Foley 2003	WOMAC pain subscale	WOMAC function subscale	No measure	No measure	Reported
Fransen 2007	WOMAC pain subscale	WOMAC function subscale	SF-12 MCS	No measure	Not reported
French 2013	Activity pain NRS	WOMAC function subscale	SF-36 MCS	No measure	Not reported
Hall 2025	Global/overall pain NRS	WOMAC function subscale	AQoL-6D	7-point Likert scale (from "much worse" to "much better")	Reported
Hopman-Rock 2000	Global/overall pain VAS	IRGL function	No measure	No measure	Not reported
Juhakoski 2011	WOMAC pain subscale	WOMAC function subscale	No measure	No measure	Reported
Krauß 2014	WOMAC pain subscale	WOMAC function subscale	No measure	No measure	Reported
Sandal 2019	HOOS pain subscale	HOOS function subscale	HOOS quality of life subscale	7-point Likert scale (from "somewhat worse" to "markedly better")	Reported

Table 3. Outcome reporting bias in trials (ORBIT) matrix (Continued)

Tak 2005	Global/overall pain VAS	GARS function	Generic quality of life	No measure	Reported
Teirlinck 2016	Global/overall pain NRS	HOOS function subscale	EQ-5D	7-point Likert scale (from "total recovery" to "worse than ever")	Reported
Thompson 2020	Global/overall pain VAS	WOMAC function subscale	No measure	"Responders" defined as participants who achieved the minimal clinically important response (≥ 13 mm on pain VAS)	Reported
van Baar 1998	Global/overall pain VAS	IRGL function	No measure	No measure	Reported
Weber 2024	HOOS pain subscale	HOOS function subscale	HOOS quality of life subscale	No measure	Reported

AQoL-6D: Six-dimension Assessment of Quality of Life; **EQ-5D:** EuroQol Five-Dimension Questionnaire; **GARS:** Groningen Activity Restriction Scale; **HOOS:** Hip Disability and Osteoarthritis Outcome Score; **IRGL:** Influence of Rheumatic Disease on General Health and Lifestyle; **NRS:** Numeric Rating Scale; **SF-12 MCS:** Medical Outcomes Study 12-Item Short-Form Health Survey Mental Component Summary; **SF-36 MCS:** Medical Outcomes Study 36-Item Short-Form Health Survey Mental Component Summary; **VAS:** visual analogue scale; **WOMAC:** Western Ontario and McMaster Universities Arthritis Index.

Table 4. Summary of adverse events

Study ID	Study arms	Adverse events reported	Nature of adverse events (n)	Withdrawal due to adverse event (n)
Abbott 2013	Exercise	Yes	None ^a	No
	Usual care	Yes	None ^a	No
Bendrik 2021	Exercise + education	Yes	Back pain (1)	No
	Education	Yes	None	No
Chopp-Hurley 2017	Exercise	No	None ^b	No
	No exercise	No	None ^b	No
de Sire 2023	Exercise + conventional rehabilitation	No	None ^d	No
	Conventional rehabilitation	No	None ^d	No
Fernandes 2010	Exercise + education	Yes	Discontinued exercise due to increased pain (1) ^c Hip replacement surgery (14) ^c	Yes (14)
	Education	Yes	Hip replacement surgery (23) ^c	Yes (23)

Table 4. Summary of adverse events (Continued)

Foley 2003	Exercise	Yes	Increased pain (2) ^a Increased blood pressure (1) ^a Undisclosed (1) ^a	Yes (4)
	No exercise	Yes	Illness (1) ^a	Yes (1)
Fransen 2007	Exercise	Yes	None ^a	No
	Waiting list	Yes	None ^a	No
French 2013	Exercise	Yes	Surgery (3) ^c	Yes (3)
	Waiting list	Yes	None ^d	No
Hall 2025	Aerobic and resistance exercise	Yes	Hip pain (12) Back pain (8) Knee pain (7) Foot pain (2) Lower limb pain (2) Hernia (2) Dizziness (1) Other (1)	No
	Resistance exercise	Yes	Hip pain (17) Back pain (9) Knee pain (7) Foot pain (2) Lower limb pain (3) Neck pain (3) Hernia (2) Skin irritation (2) Other (1)	Yes (1)
Hopman-Rock 2000	Exercise	No	None ^b	No
	No exercise	No	None ^b	No
Juhakoski 2011	Exercise + GP care	No	Neck pain (1) ^c Hip replacement surgery (5) ^c Other disease (2) ^c	Yes (3)
	GP care	No	Hip replacement surgery (8) ^c Death (2) ^c	Yes (2)
Krauß 2014	Exercise	No	Surgery (1) ^c	Unclear
	Usual care	No	None ^d	No
	Placebo ultrasound	No	None ^d	No
Sandal 2019	Exercise	Yes	None	No
	Waiting list	Yes	None	No
Tak 2005	Exercise	Yes	Increased back pain (2) ^a	Yes (2)

Table 4. Summary of adverse events *(Continued)*

	No exercise	Yes	None ^a	No
Teirlinck 2016	Exercise + GP care	Yes	Hip replacement surgery (6) ^c	No
	GP care	Yes	Hip replacement surgery (9) ^c	No
Thompson 2020	Exercise	Yes	None	No
	Waiting list	Yes	None	No
van Baar 1998	Exercise + GP care	Yes	Undisclosed (1) ^a	Unclear
	GP care	Yes	None ^a	No
Weber 2024	Exercise	Yes	None	No
	Usual care	Yes	None	No

^a Based on previous review.

^b No study flow diagram.

^c Extracted from study flow diagram.

^d None described in study flow diagram.

GP: general practitioner; **n:** number of participants.

Table 5. Sensitivity analysis

Comparator	Outcome	Mean difference (95% confidence interval)					
		All studies (primary analysis)	Studies at low risk of selection bias	Studies at low risk of detection bias	Studies at low risk of attrition bias	Removal of outliers	Removal of studies using change scores
Attention control/placebo	Pain	-6.31 (-12.98 to 0.35)	-6.31 (-12.98 to 0.35) ^a	No studies available	-6.31 (-12.98 to 0.35) ^a	Not applicable	Not applicable
	Physical function	-7.44 (-13.86 to -1.01)	-7.44 (-13.86 to -1.01) ^a	No studies available	-7.44 (-13.86 to -1.01) ^a	Not applicable	Not applicable
	Quality of life	No studies available	No studies available	No studies available	No studies available	Not applicable	Not applicable
Treatment/usual care/limited education	Pain	-7.19 (-10.70 to -3.68)	-8.59 (-13.15 to -3.86)	No studies available	-7.72 (-11.92 to -3.51)	-6.66 (-10.17 to -3.16)	-7.19 (-11.05 to -3.51)
	Physical function	-8.79 (-12.00 to -5.41)	-8.11 (-12.50 to -3.55)	No studies available	-8.62 (-12.67 to -4.39)	Not applicable	-8.11 (-11.66 to -4.56)
	Quality of life	2.31 (-1.15 to 5.91)	1.30 (-3.46 to 6.20)	No studies available	1.30 (-3.46 to 6.20)	Not applicable	Not applicable
Co-intervention	Pain	-3.86 (-8.07 to 0.35)	-3.51 (-7.54 to 0.53)	-9.82 (-25.95 to -6.49)	-3.86 (-8.07 to 0.35) ^a	-2.98 (-6.66 to 0.88)	Not applicable
	Physical function	-2.37 (-6.59 to 1.86)	-1.35 (-5.24 to 2.53)	-11.49 (-36.64 to 11.83)	-2.37 (-6.59 to 1.86) ^a	-1.35 (-4.73 to 2.03)	-3.38 (-7.60 to 0.68)
	Quality of life	3.60 (-1.30 to 8.36)	1.73 (-1.01 to 4.61)	10.67 (-8.79 to 30.28)	3.60 (-1.30 to 8.36) ^a	Not applicable	Not applicable

^a same studies as primary analysis.